

## What is Matplotlib?

- Matplotlib is a low level graph plotting library in python that serves as a visualization utility.
- Matplotlib was created by **John D. Hunter**.
- Matplotlib is open source and we can use it freely.
- Matplotlib is mostly written in python, a few segments are written in C, Objective-C and Javascript for Platform compatibility.

## Installation of Matplotlib

```
pip install matplotlib
```

## Checking Matplotlib Version

```
import matplotlib

print(matplotlib.__version__) # Output:- 2.0.0
```

## Import Matplotlib

```
Importing Matplotlib
import matplotlib.pyplot as plt
OR
from matplotlib import pyplot as plt
import matplotlib.pyplot as plt
```

## Types of Matplotlib

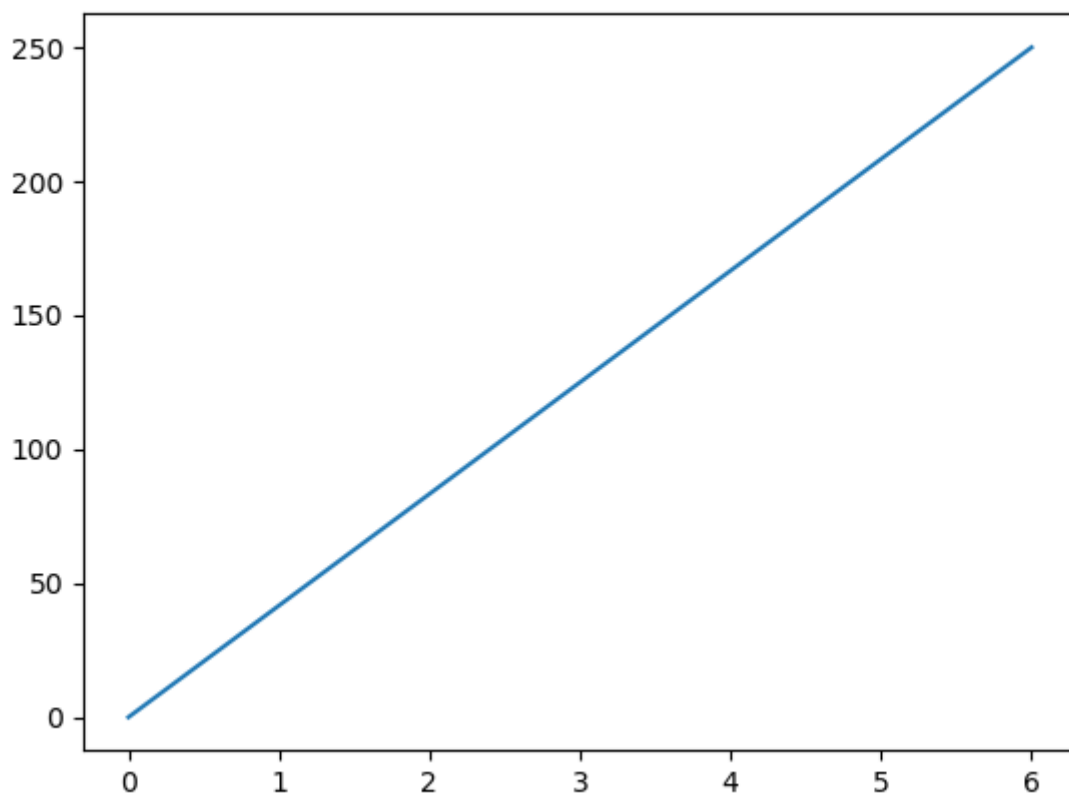
1. Liner Plot
2. Scatter Plot
3. Bar Plot
4. Stem Plot
5. Step Plot
6. Hist Plot
7. Box Plot
8. Pie Plot
9. Fill\_between Plot

## Pyplot Example

```
# Exmaple 1
import matplotlib.pyplot as plt
import numpy as np

xpoints = np.array([0, 6])
ypoints = np.array([0, 250])

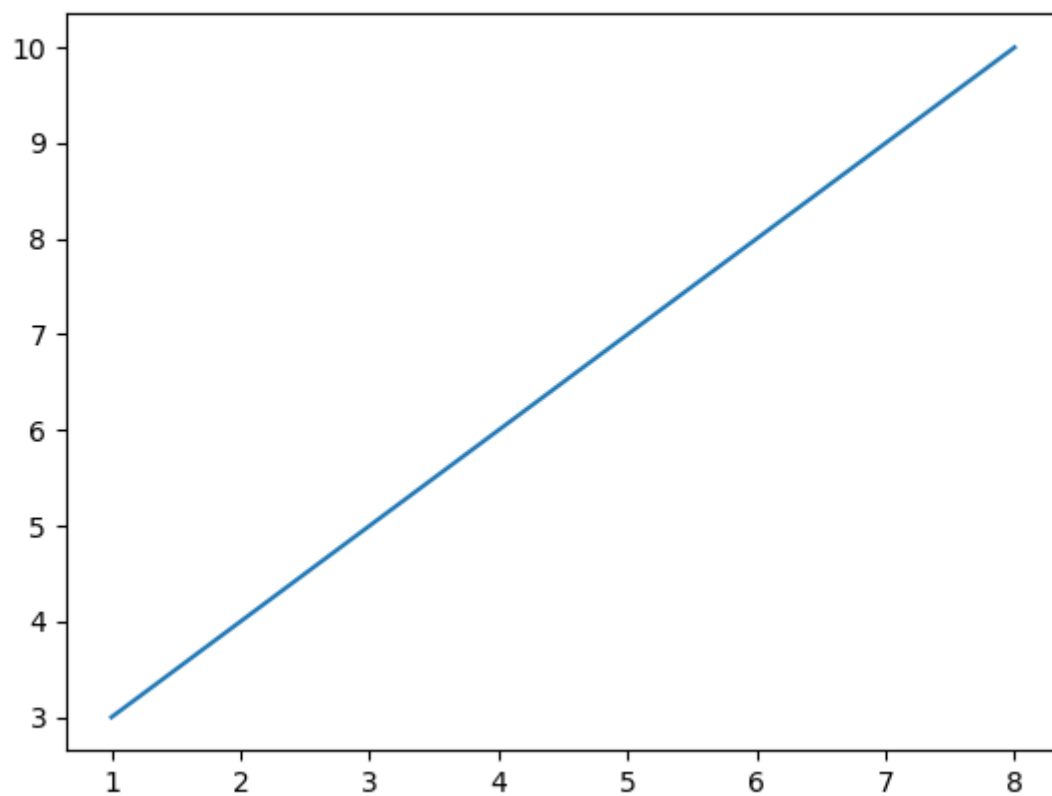
plt.plot(xpoints, ypoints)
plt.show()
```



```
# Example 2
import matplotlib.pyplot as plt
import numpy as np

xpoints = np.array([1, 8])
ypoints = np.array([3, 10])

plt.plot(xpoints, ypoints)
plt.show()
```

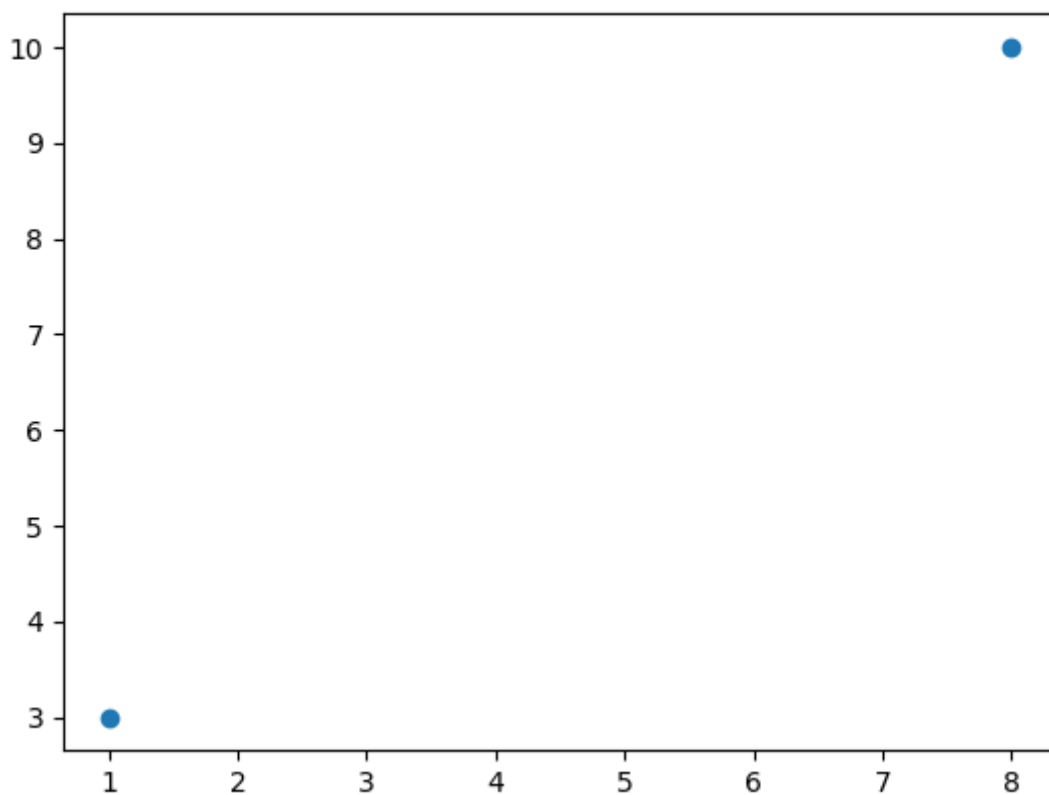


## Plotting Without Line Example

```
import matplotlib.pyplot as plt
import numpy as np

xpoints = np.array([1, 8])
ypoints = np.array([3, 10])

plt.plot(xpoints, ypoints, 'o')
plt.show()
```



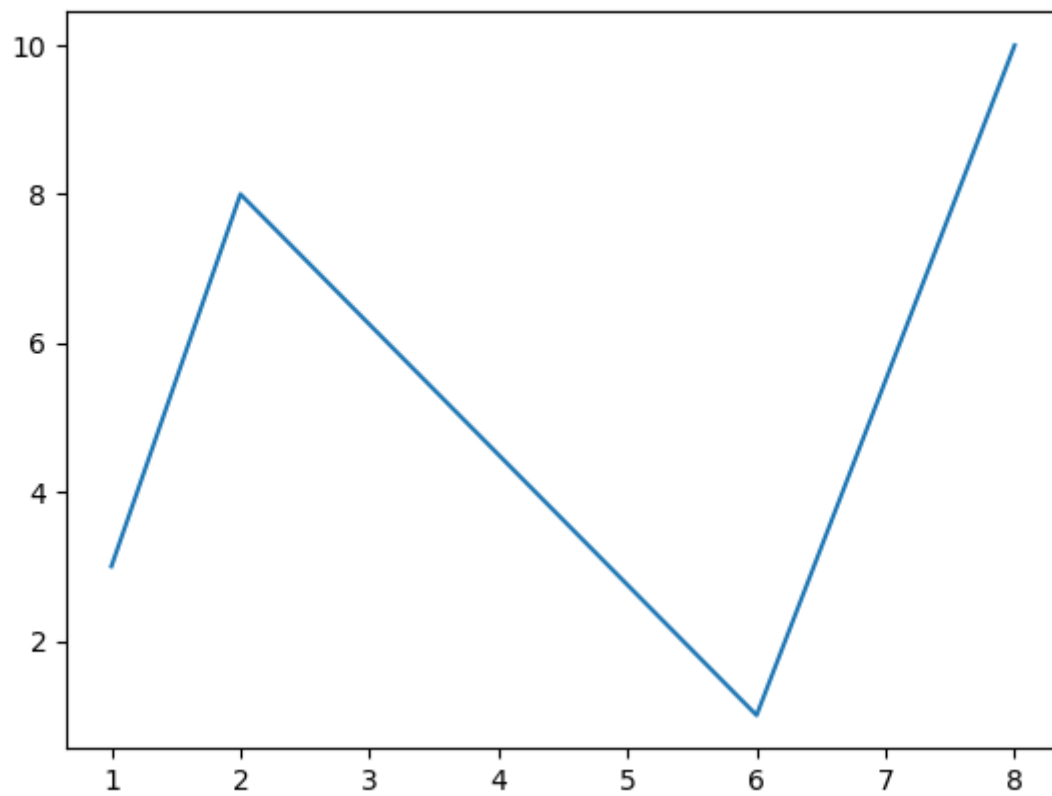
## Multiple Points

- Draw a line in a diagram from position (1, 3) to (2, 8) then to (6, 1) and finally to position (8, 10):

```
import matplotlib.pyplot as plt
import numpy as np

xpoints = np.array([1, 2, 6, 8])
ypoints = np.array([3, 8, 1, 10])

plt.plot(xpoints, ypoints)
plt.show()
```

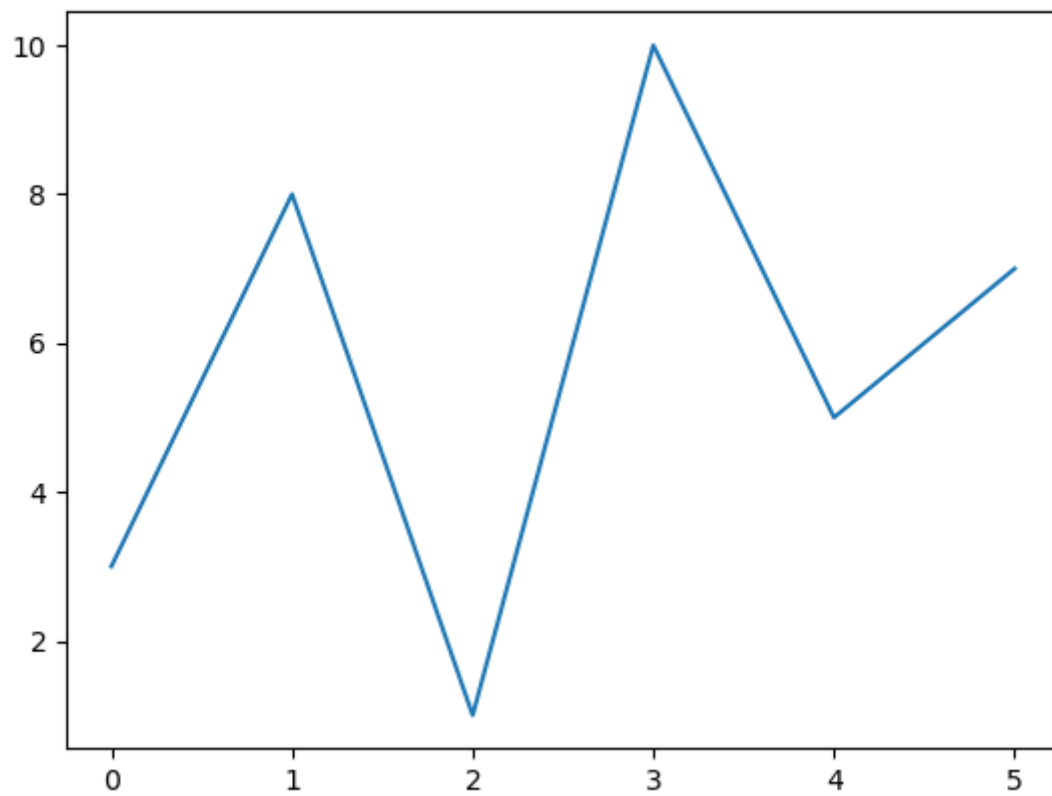


## Default X-Points

```
import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10, 5, 7])

plt.plot(ypoints)
plt.show()
```

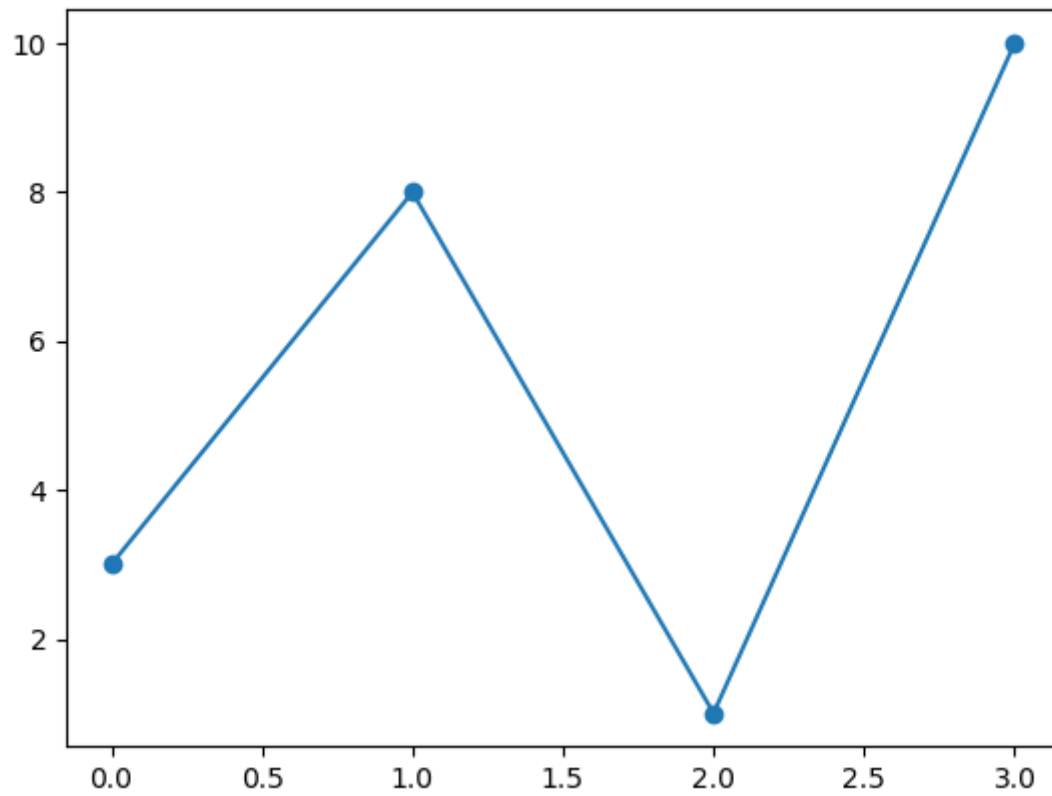


## Matplotlib Markers

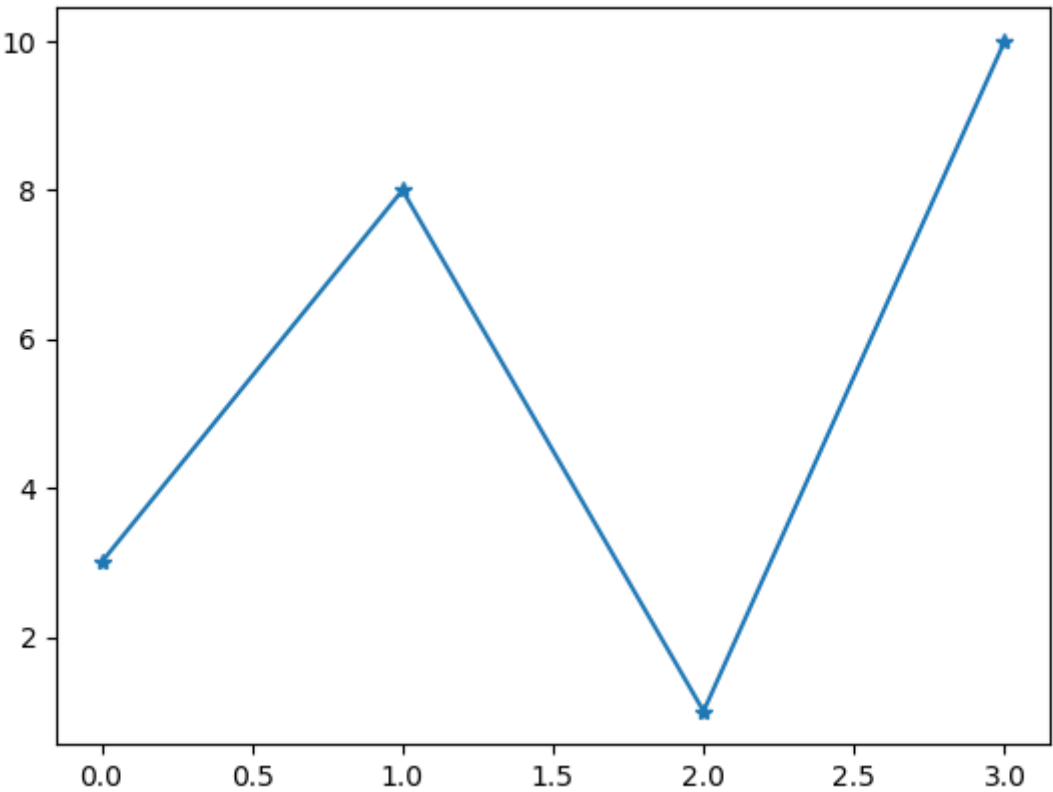
```
import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, marker = 'o')
plt.show()
```



```
plt.plot(ypoints, marker = '*')  
plt.show()
```



Marker Reference

| Marker | Description    |
|--------|----------------|
| 'o'    | Circle         |
| '*'    | Star           |
| '.'    | Point          |
| ','    | Pixel          |
| 'x'    | X              |
| 'X'    | X (filled)     |
| '+'    | Plus           |
| 'P'    | Plus (filled)  |
| 's'    | Square         |
| 'D'    | Diamond        |
| 'd'    | Diamond (thin) |
| 'p'    | Pentagon       |
| 'H'    | Hexagon        |

| Marker | Description    |
|--------|----------------|
| 'h'    | Hexagon        |
| 'v'    | Triangle Down  |
| '^'    | Triangle Up    |
| '<'    | Triangle Left  |
| '>'    | Triangle Right |
| '1'    | Tri Down       |
| '2'    | Tri Up         |
| '3'    | Tri Left       |
| '4'    | Tri Right      |
| ' '    | Vline          |
| '_'    | Hline          |

Format Strings **fmt**

- You can also use the shortcut string notation parameter to specify the marker.
- This parameter is also called **fmt**, and is written with this **syntax**: marker|line|color

Line Reference

| Line Syntax | Description         |
|-------------|---------------------|
| '-'         | Solid line(default) |
| '.'         | Dotted line         |
| '--'        | Dashed line         |
| '-.'        | Dashed/dotted line  |

Color Reference

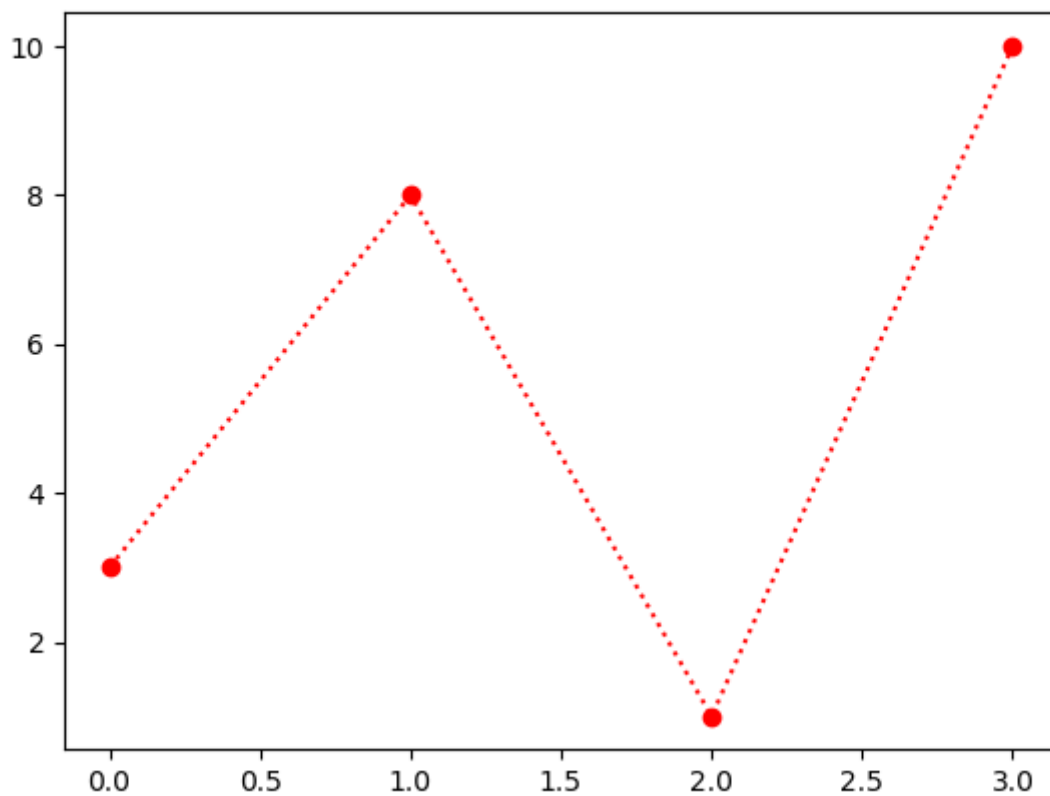
| Color Syntax | Description |
|--------------|-------------|
| 'r'          | Red         |
| 'g'          | Green       |
| 'b'          | Blue        |
| 'c'          | Cyan        |
| 'm'          | Magenta     |
| 'y'          | Yellow      |

| Color Syntax | Description |
|--------------|-------------|
| 'k'          | Black       |
| 'w'          | White       |

```
import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, 'o:r')    # o means circle / r means red
plt.show()
```



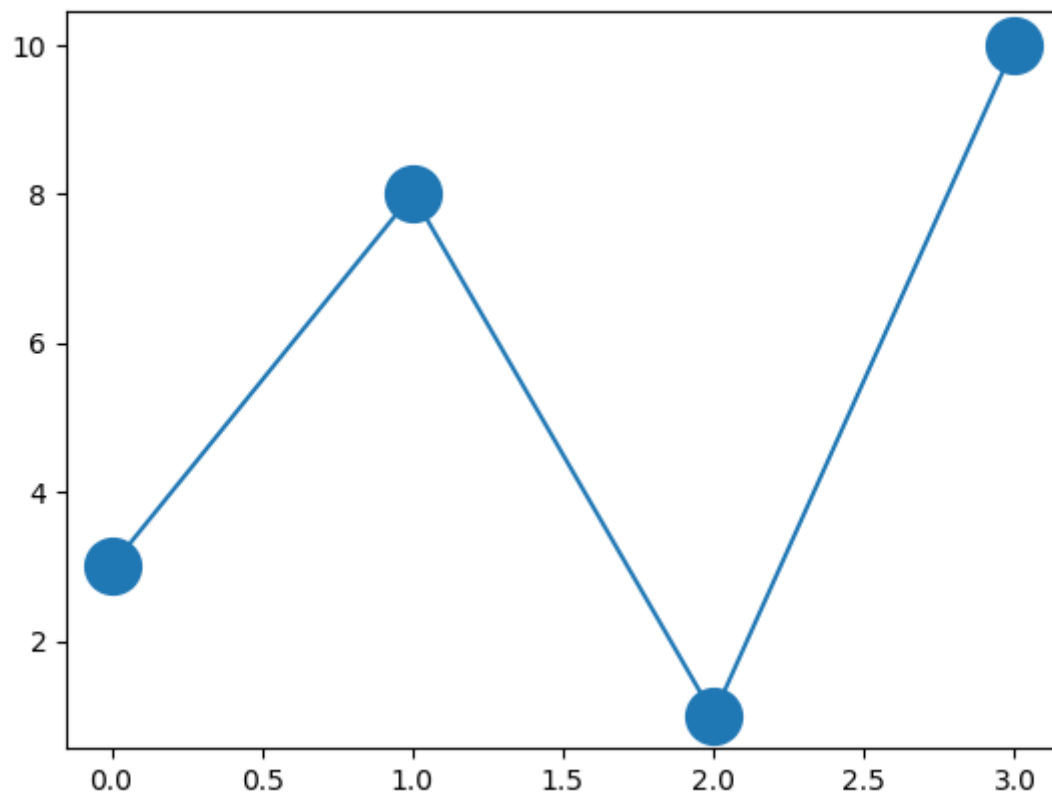
## Marker Size

- You can use the keyword argument **markersize** or the shorter version, **ms** to set the size of the markers:

```
import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])
```

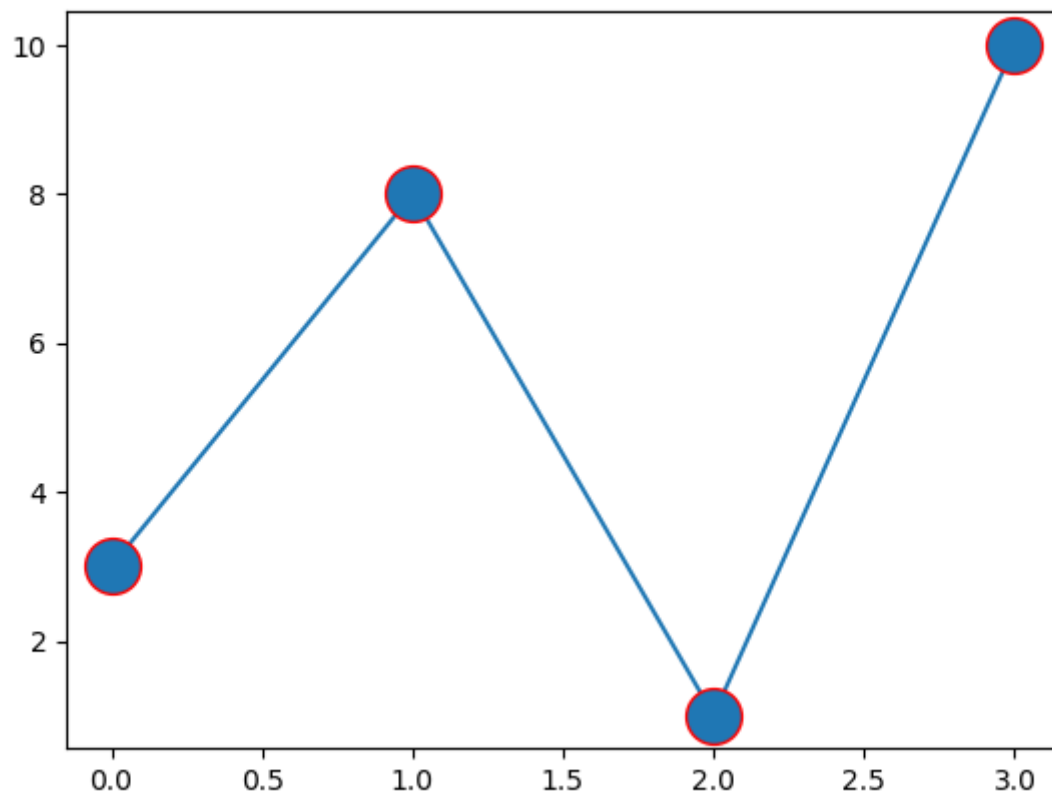
```
plt.plot(ypoints, marker = 'o', ms = 20)  
plt.show()
```



## Marker Color

- You can use the keyword argument **markeredgecolor** or the shorter **mec** to set the color of the edge of the markers:

```
import matplotlib.pyplot as plt  
import numpy as np  
  
ypoints = np.array([3, 8, 1, 10])  
  
plt.plot(ypoints, marker = 'o', ms = 20, mec = 'r')  
plt.show()
```

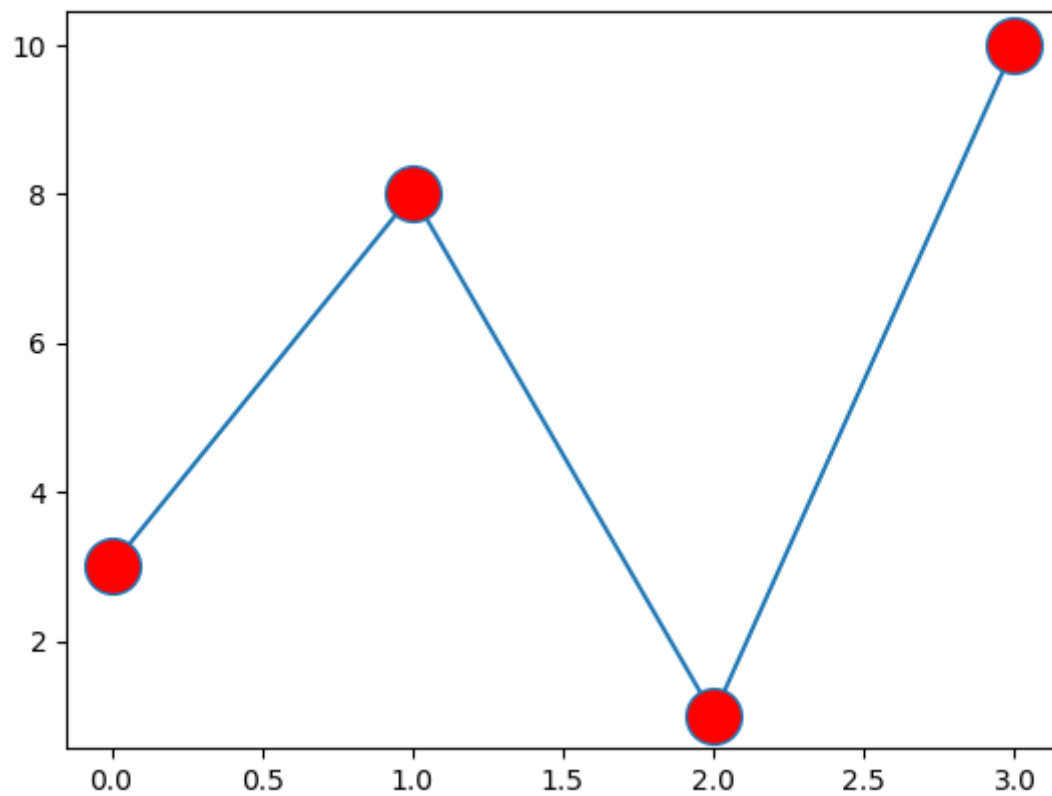


- You can use the keyword argument **markerfacecolor** or the shorter **mfc** to set the color inside the edge of the markers:

```
import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, marker = 'o', ms = 20, mfc = 'r')
plt.show()
```

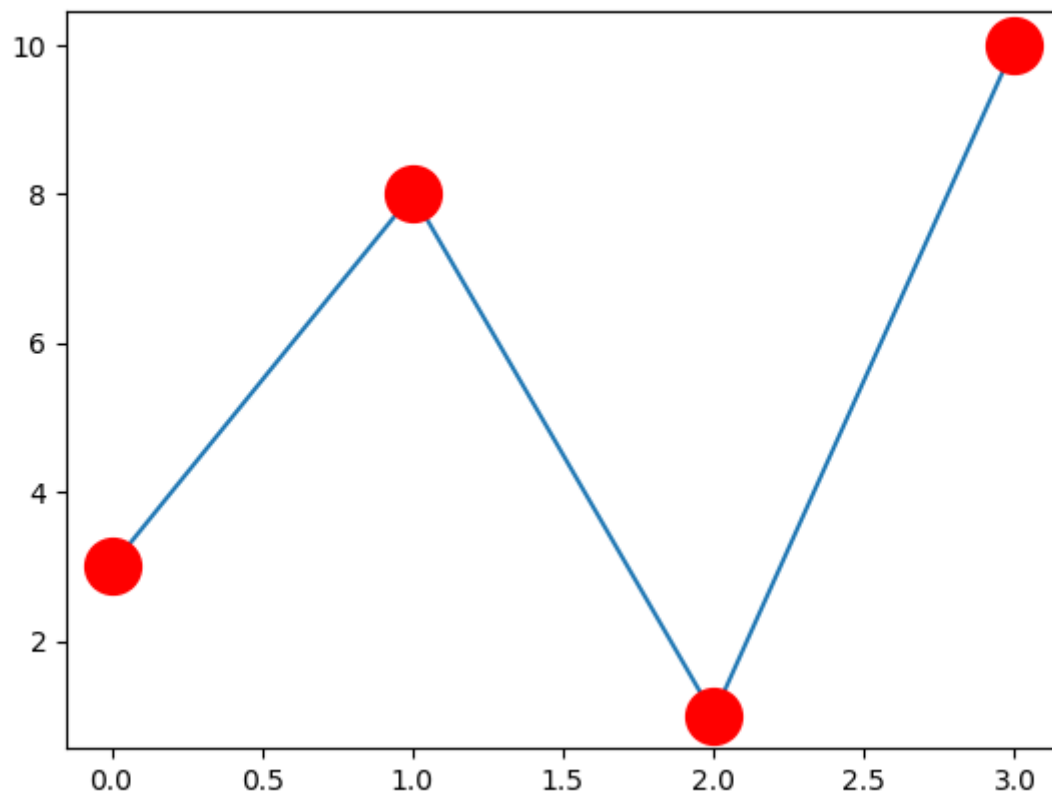


- Use **both** the **mec** and **mfc** arguments to color the entire marker:

```
import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, marker = 'o', ms = 20, mec = 'r', mfc = 'r')
plt.show()
```

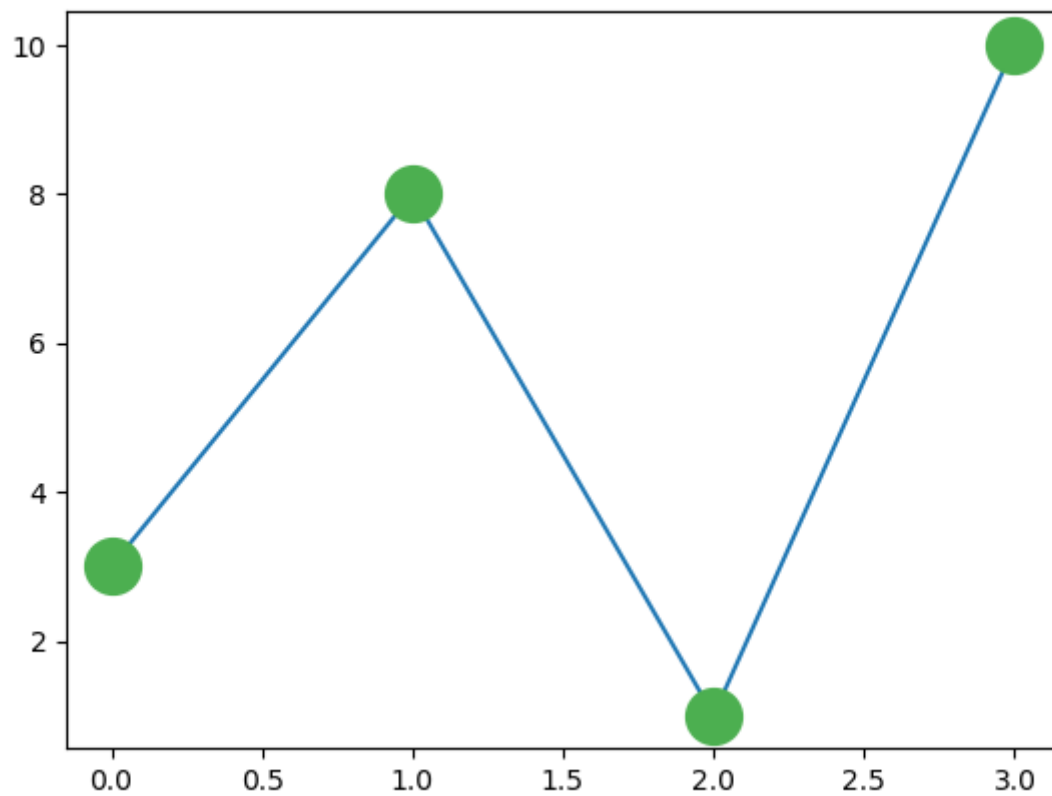


- We can also **use Hexadecimal color** values:

```
import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, marker = 'o', ms = 20, mec = '#4CAF50', mfc = '#4CAF50')
plt.show()
```



- We can also **use color names** values: ([https://www.w3schools.com/colors/colors\\_names.asp](https://www.w3schools.com/colors/colors_names.asp))

```
plt.plot(ypoints, marker = 'o', ms = 20, mec = 'hotpink', mfc = 'hotpink')
```

# Matplotlib Line

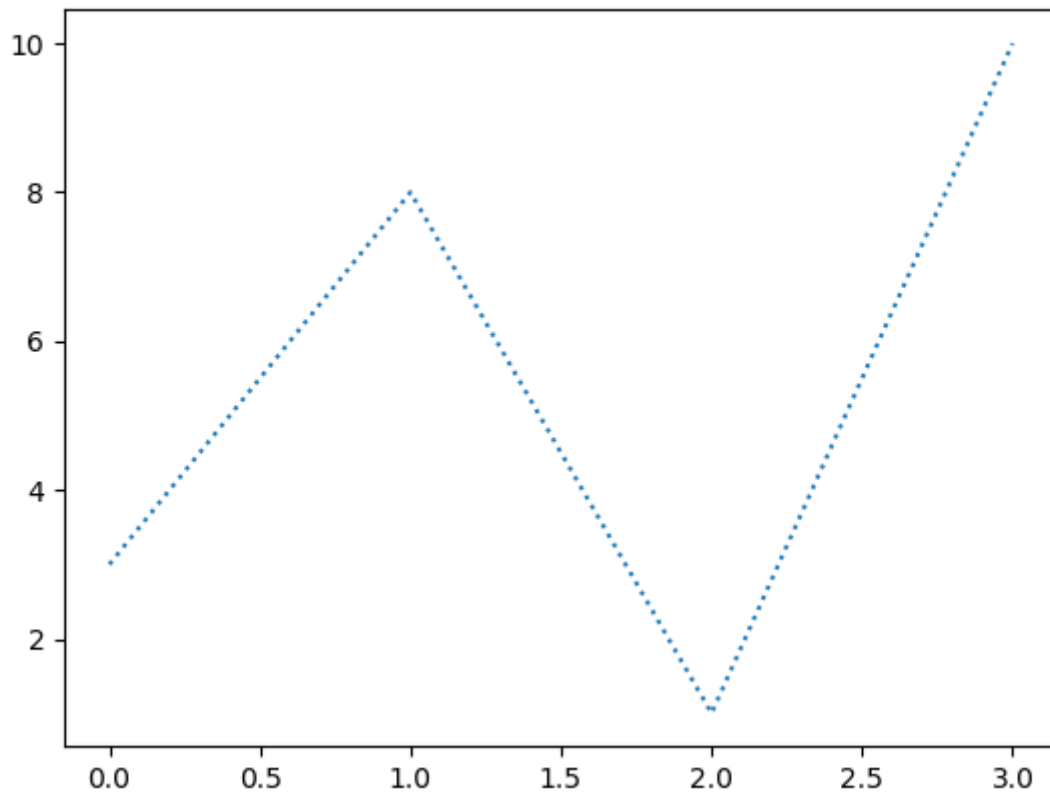
---

- You can use the keyword argument **linestyle**, or shorter **ls**, to change the style of the plotted line.

| Line Syntax | Description          |
|-------------|----------------------|
| '-'         | Solid line(default)  |
| '.'         | Dotted line          |
| '--'        | Dashed line          |
| '-.'        | Dashed/dotted line   |
| " or ''     | 'None' (Blank Graph) |

```
# dotted can be written as :.  
plt.plot(ypoints, linestyle = 'dotted') / plt.plot(ypoints, linestyle = ':')  
plt.plot(ypoints, ls = 'dotted') / plt.plot(ypoints, ls = ':')  
  
# and so on
```

```
import matplotlib.pyplot as plt  
import numpy as np  
  
ypoints = np.array([3, 8, 1, 10])  
  
plt.plot(ypoints, linestyle = 'dotted')  
plt.show()
```



## Line Color

---

- You can use the keyword argument **color** or the shorter **c** to set the color of the line:

```
import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, color = 'r') # using color
plt.plot(ypoints, color = '#4CAF50') # using Hexadecimal color
plt.plot(ypoints, color = 'hotpink') # using color name

plt.show()
```

# Line Width

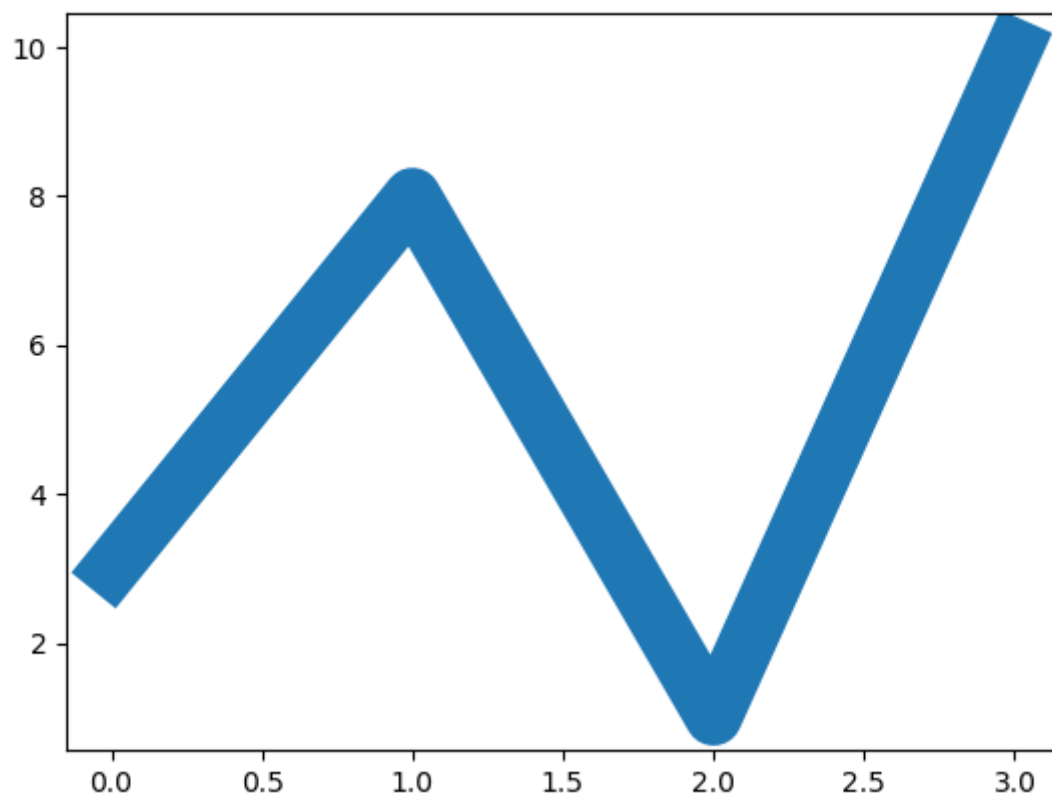
---

- You can use the keyword argument **linewidth** or the shorter **lw** to change the width of the line.

```
import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, linewidth = '20.5')
plt.show()
```



# Multiple Lines

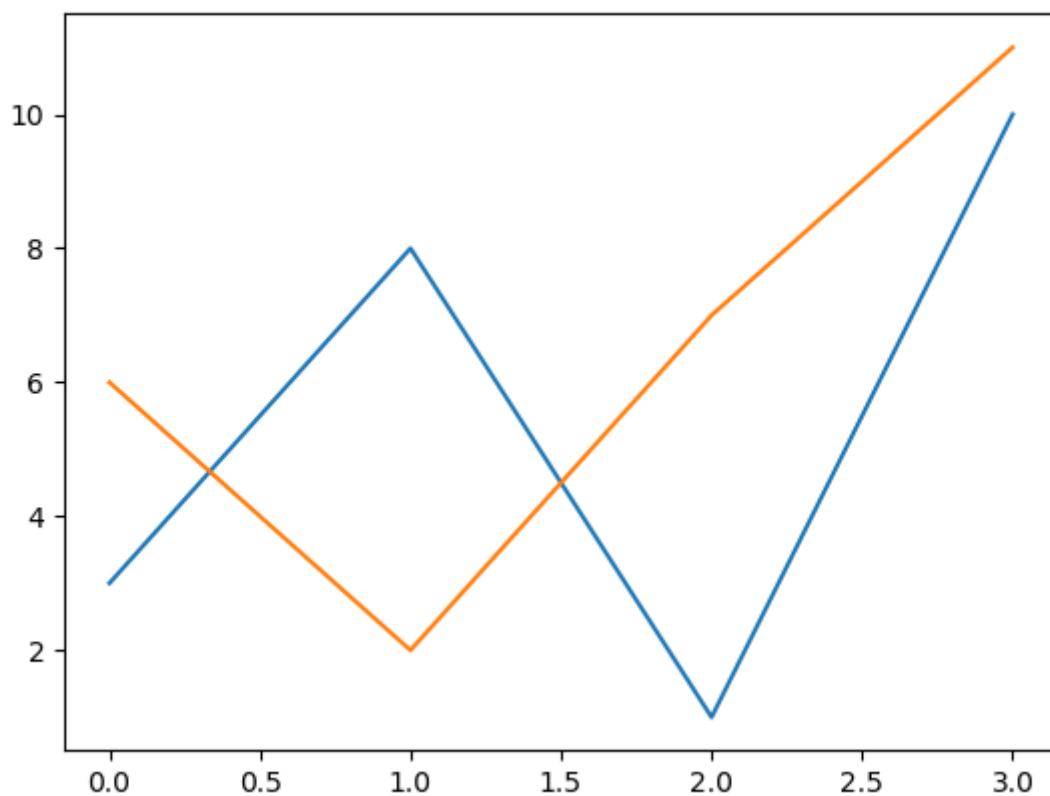
- You can plot as many lines as you like by simply adding more **plt.plot()** functions:

```
import matplotlib.pyplot as plt
import numpy as np

y1 = np.array([3, 8, 1, 10])
y2 = np.array([6, 2, 7, 11])

plt.plot(y1)
plt.plot(y2)

plt.show()
```



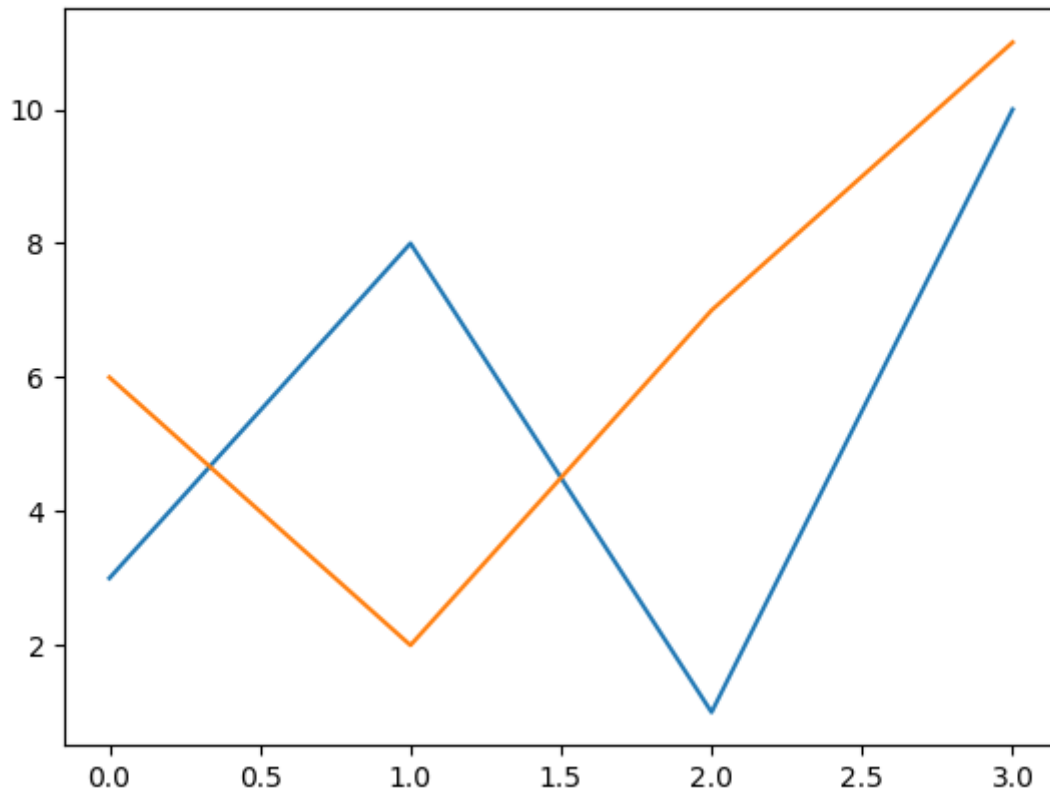
- You can also plot many lines by adding the points for the x- and y-axis for each line in the same **plt.plot()** function.

```
import matplotlib.pyplot as plt
import numpy as np

x1 = np.array([0, 1, 2, 3])
y1 = np.array([3, 8, 1, 10])
```

```
x2 = np.array([0, 1, 2, 3])
y2 = np.array([6, 2, 7, 11])

plt.plot(x1, y1, x2, y2)
plt.show()
```



## Matplotlib Labels and Title

---

- we can use the **xlabel()** and **ylabel()** functions to **set** a label for the x- and y-axis.
- we can use the **title()** function to **set** a **title** for the plot.

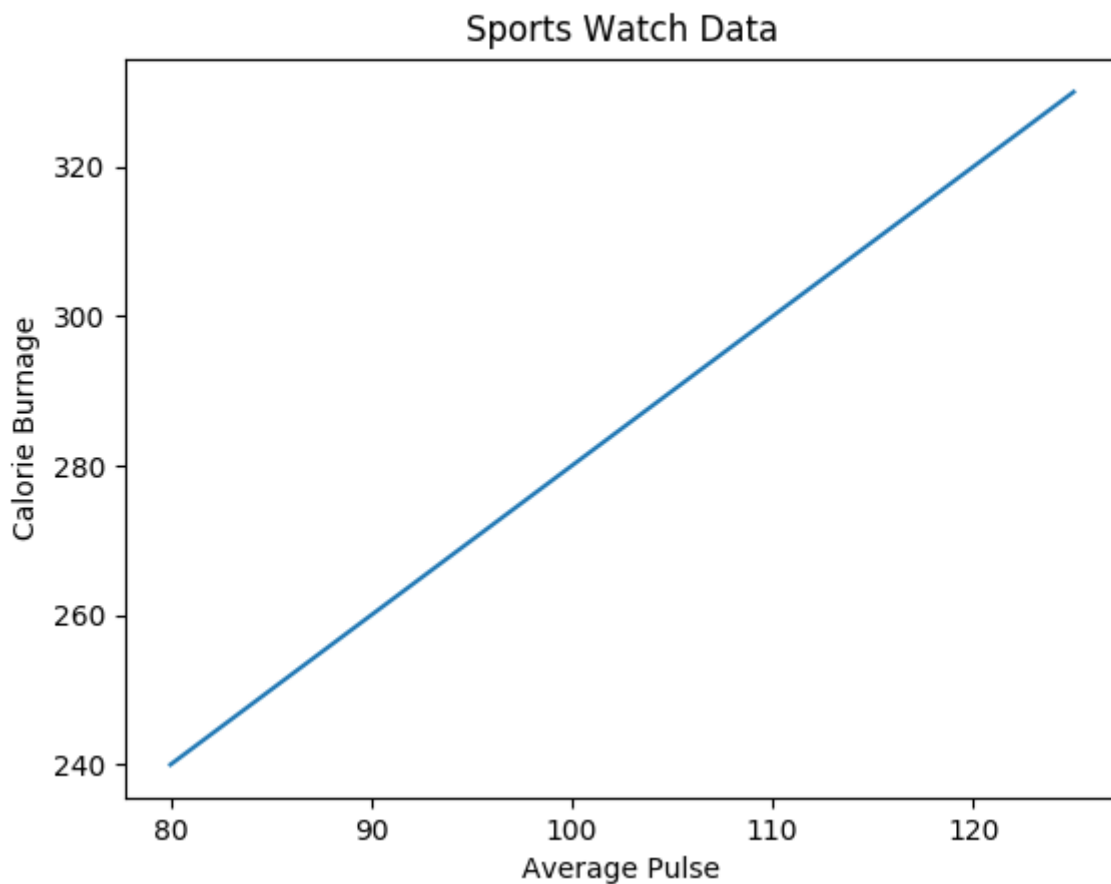
```
import numpy as np
import matplotlib.pyplot as plt

x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])

plt.plot(x, y)

plt.title("Sports Watch Data")
plt.xlabel("Average Pulse")
plt.ylabel("Calorie Burnage")

plt.show()
```



## Set Font Properties for Title and Labels

- You can use the fontdict parameter in xlabel(), ylabel(), and title() to set font properties for the title and labels.

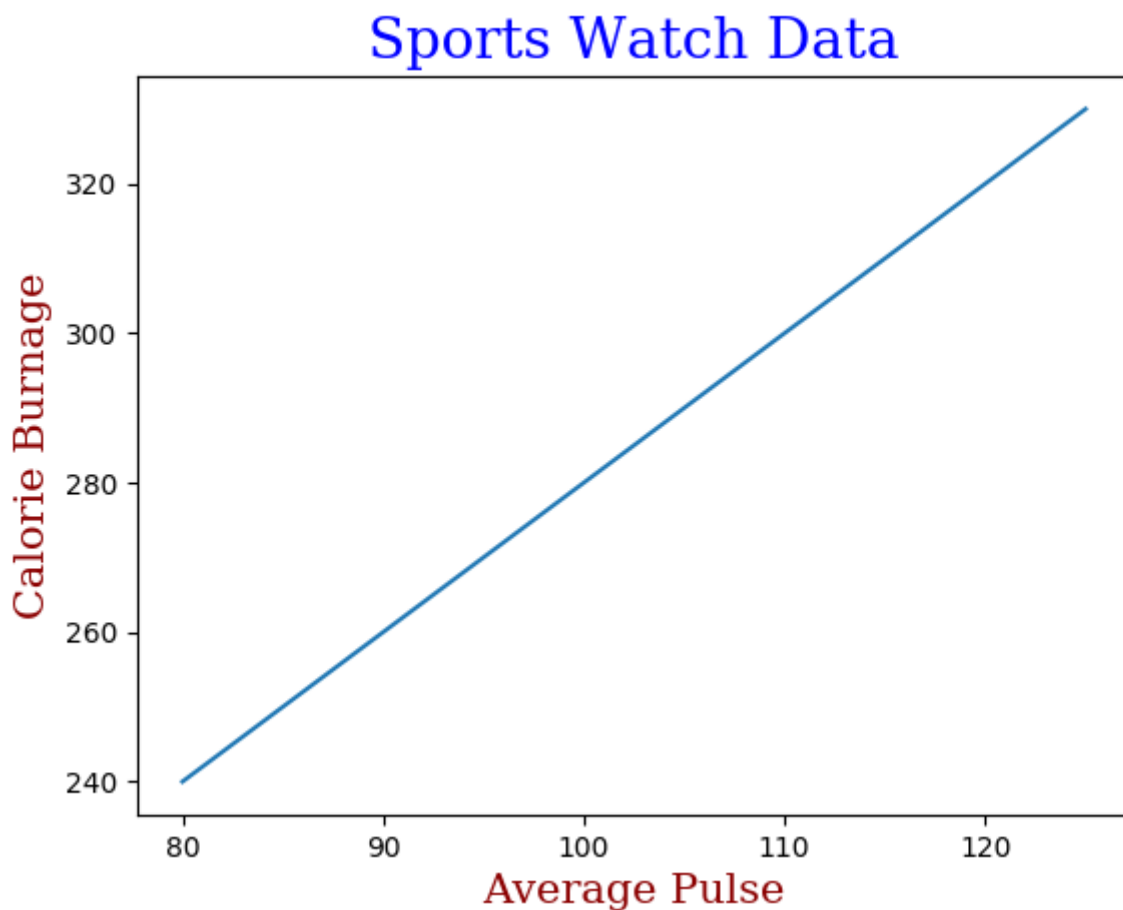
```
import numpy as np
import matplotlib.pyplot as plt

x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])

font1 = {'family':'serif','color':'blue','size':20}
font2 = {'family':'serif','color':'darkred','size':15}

plt.title("Sports Watch Data", fontdict = font1)
plt.xlabel("Average Pulse", fontdict = font2)
plt.ylabel("Calorie Burnage", fontdict = font2)

plt.plot(x, y)
plt.show()
```



### Position the Title

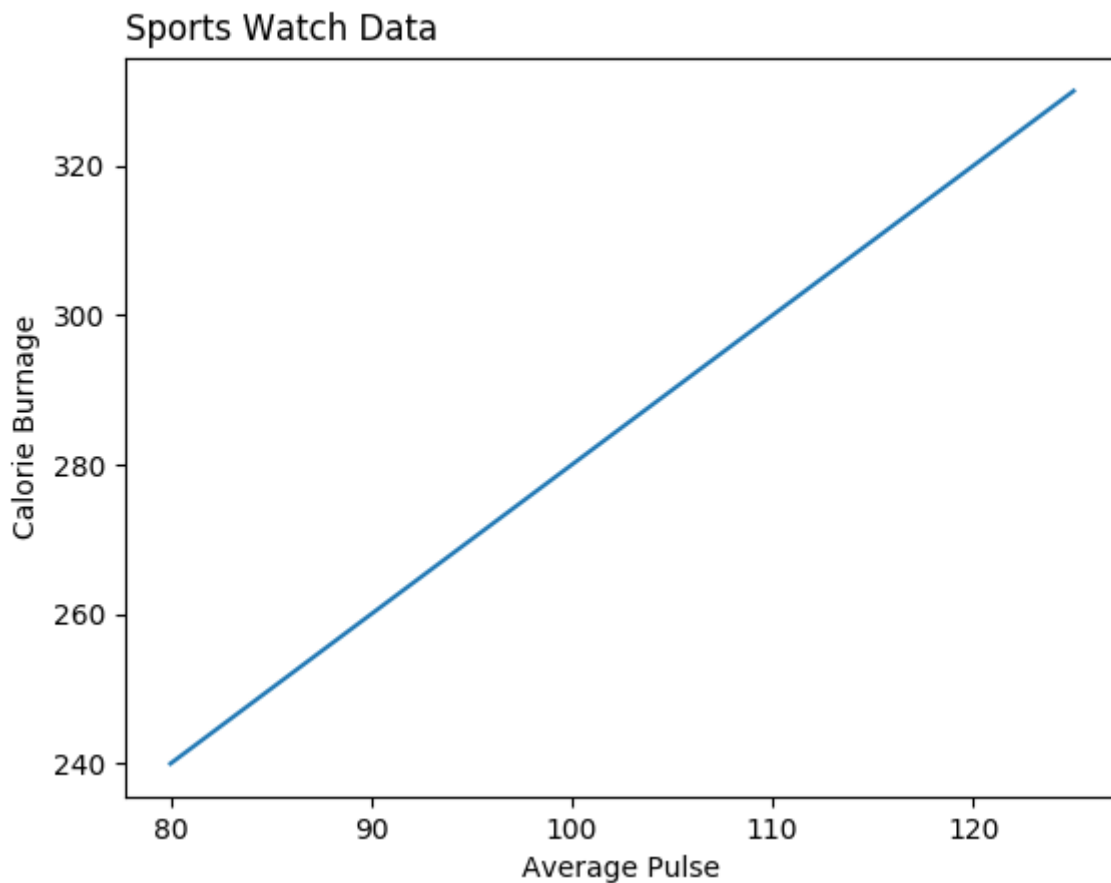
- You can use the **loc** parameter in `title()` to position the title.
- Legal values are: **'left'**, **'right'**, and **'center'**. Default value is **'center'**.

```
import numpy as np
import matplotlib.pyplot as plt

x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])

plt.title("Sports Watch Data", loc = 'left')
plt.xlabel("Average Pulse")
plt.ylabel("Calorie Burnage")

plt.plot(x, y)
plt.show()
```



## Matplotlib Adding Grid Lines

---

- We can use the `grid()` function to add grid lines to the plot.

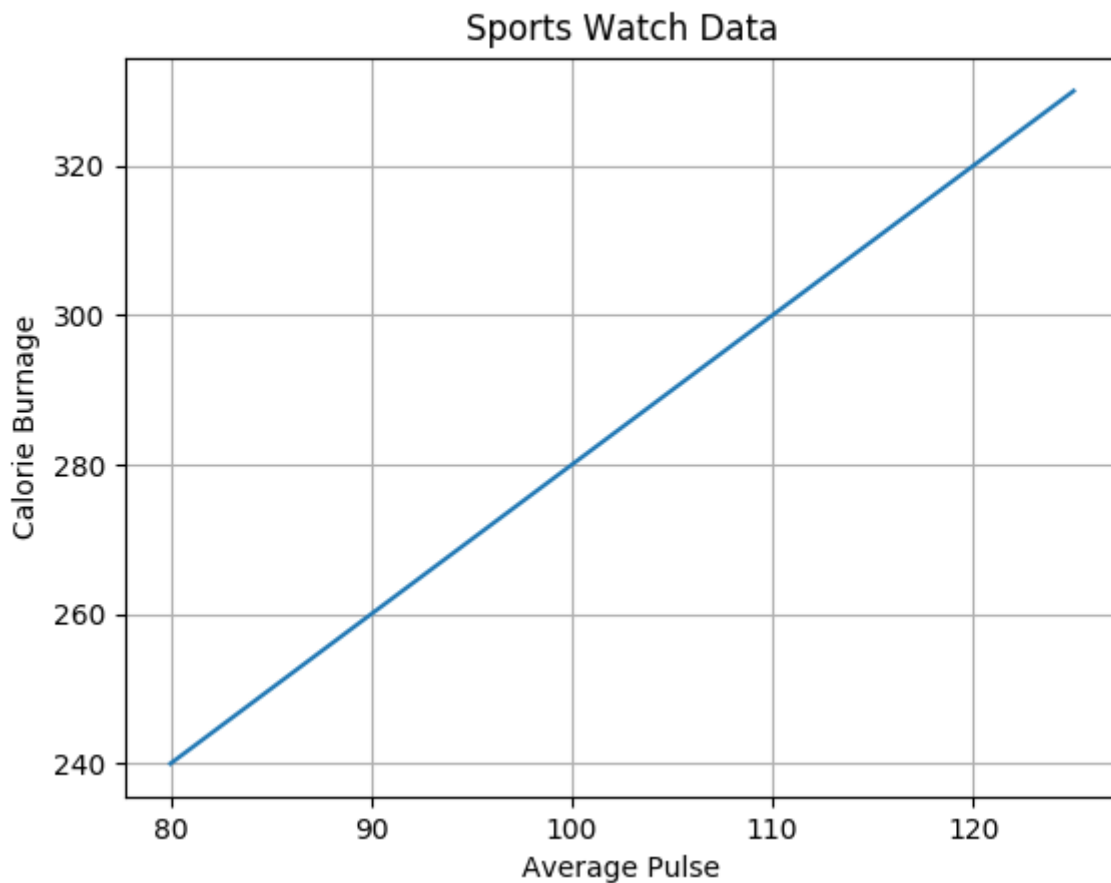
```
import numpy as np
import matplotlib.pyplot as plt

x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])

plt.title("Sports Watch Data")
plt.xlabel("Average Pulse")
plt.ylabel("Calorie Burnage")

plt.plot(x, y)
plt.grid()

plt.show()
```



## Specify Which Grid Lines to Display

```
plt.plot(x, y)
plt.grid(axis = 'x')

# Display only grid lines for the y-axis:
plt.plot(x, y)
plt.grid(axis = 'y')
```

- Set **Line Properties** for the **Grid**
  - we can also set the line properties of the grid, like this: `grid(color = 'color', linestyle = 'linestyle', linewidth = number)`.

```
plt.plot(x, y)
plt.grid(color = 'green', linestyle = '--', linewidth = 0.5)
```

## Matplotlib Subplot

### Display Multiple Plots

- With the **subplot() function** you can draw multiple plots in one figure:

- The subplot() function takes **three** arguments that describes the layout of the figure.
  - The layout is organized in **rows** and **columns**, which are represented by the **first** and **second** argument.
  - The **third** argument represents the index of the **current plot**.

```
plt.subplot(1, 2, 1)
```

```
import matplotlib.pyplot as plt
import numpy as np

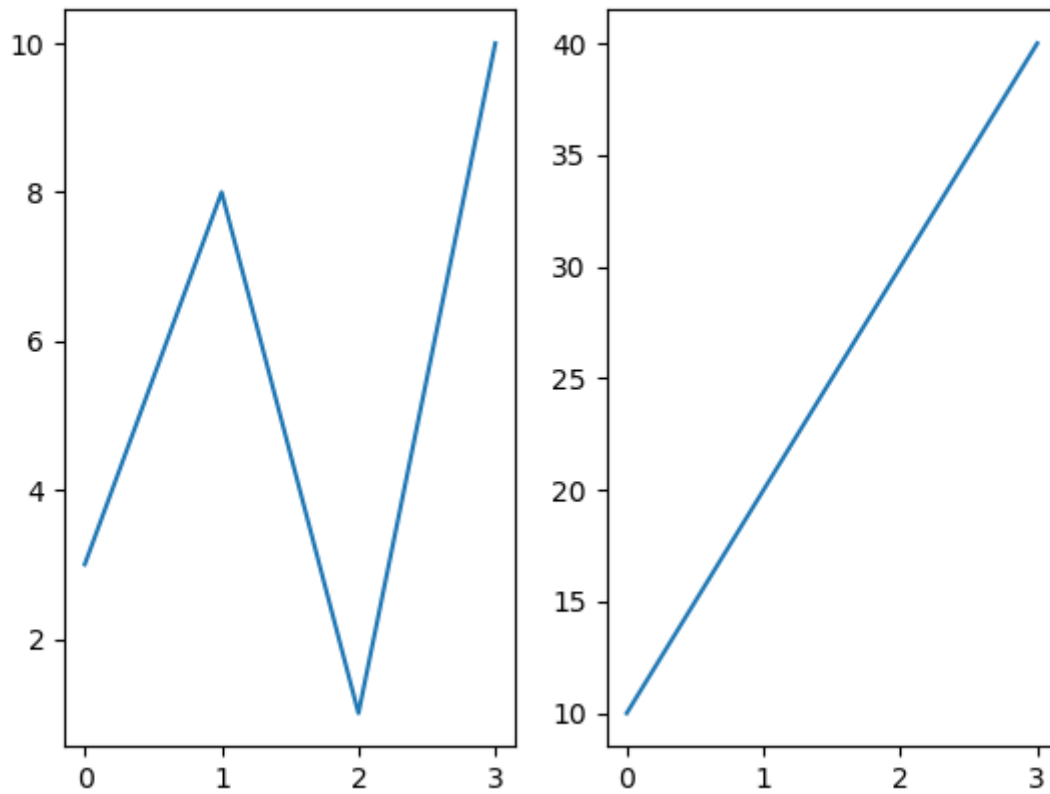
#plot 1:
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])

plt.subplot(1, 2, 1)
plt.plot(x,y)

#plot 2:
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])

plt.subplot(1, 2, 2)
plt.plot(x,y)

plt.show()
```



```
import matplotlib.pyplot as plt
import numpy as np

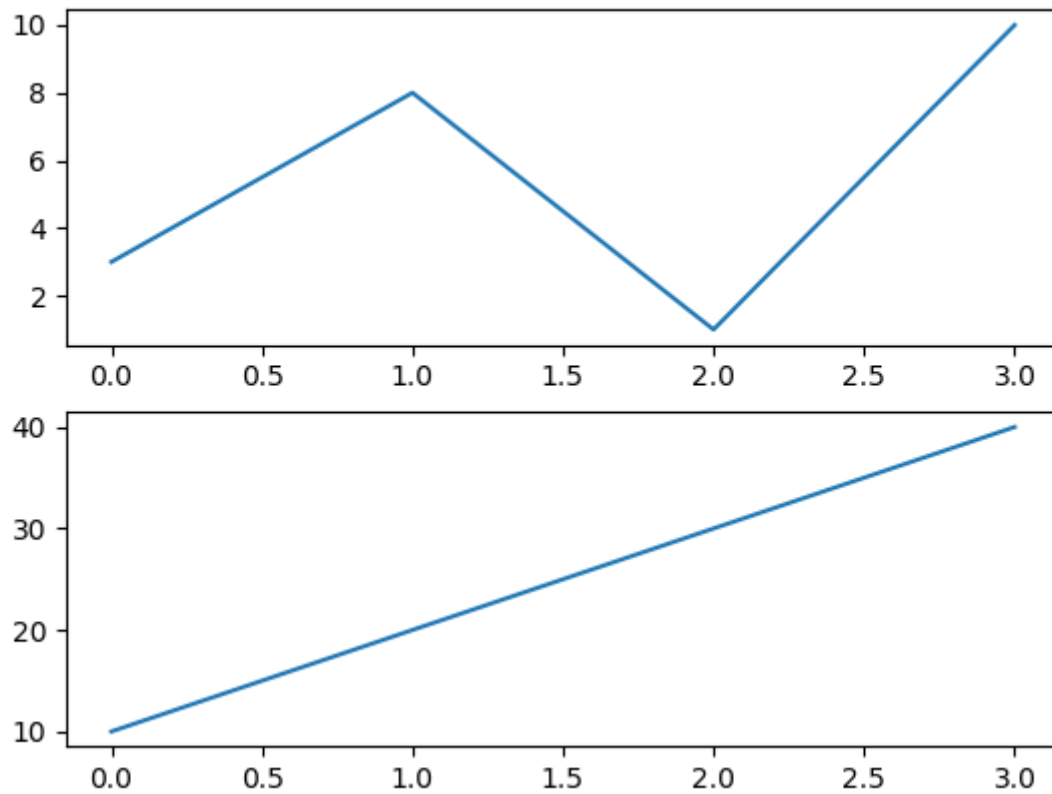
#plot 1:
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])

plt.subplot(2, 1, 1)
plt.plot(x,y)

#plot 2:
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])

plt.subplot(2, 1, 2)
plt.plot(x,y)

plt.show()
```



```
import matplotlib.pyplot as plt
import numpy as np
```

```
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(2, 3, 1)
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(2, 3, 2)
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])
```

```
plt.subplot(2, 3, 3)
plt.plot(x,y)
```

```
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])
```

```
plt.subplot(2, 3, 4)
plt.plot(x,y)
```

```

x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])

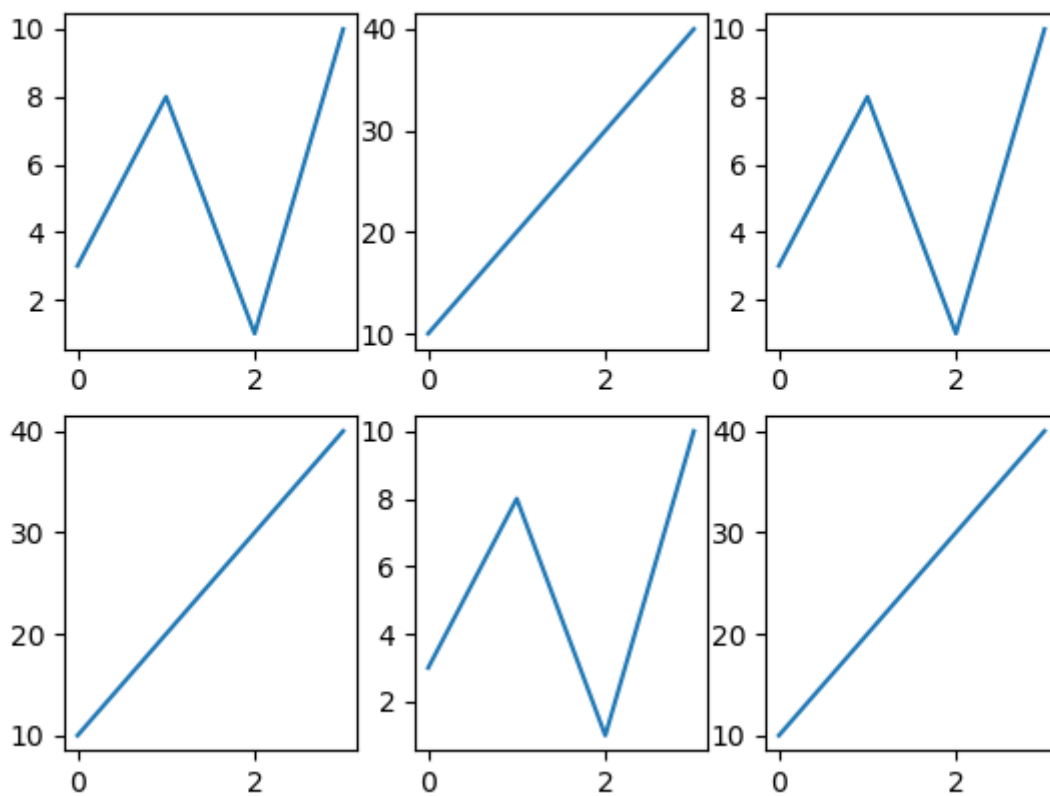
plt.subplot(2, 3, 5)
plt.plot(x,y)

x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])

plt.subplot(2, 3, 6)
plt.plot(x,y)

plt.show()

```



## Subplot Title

```

import matplotlib.pyplot as plt
import numpy as np

#plot 1:
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])

plt.subplot(1, 2, 1)

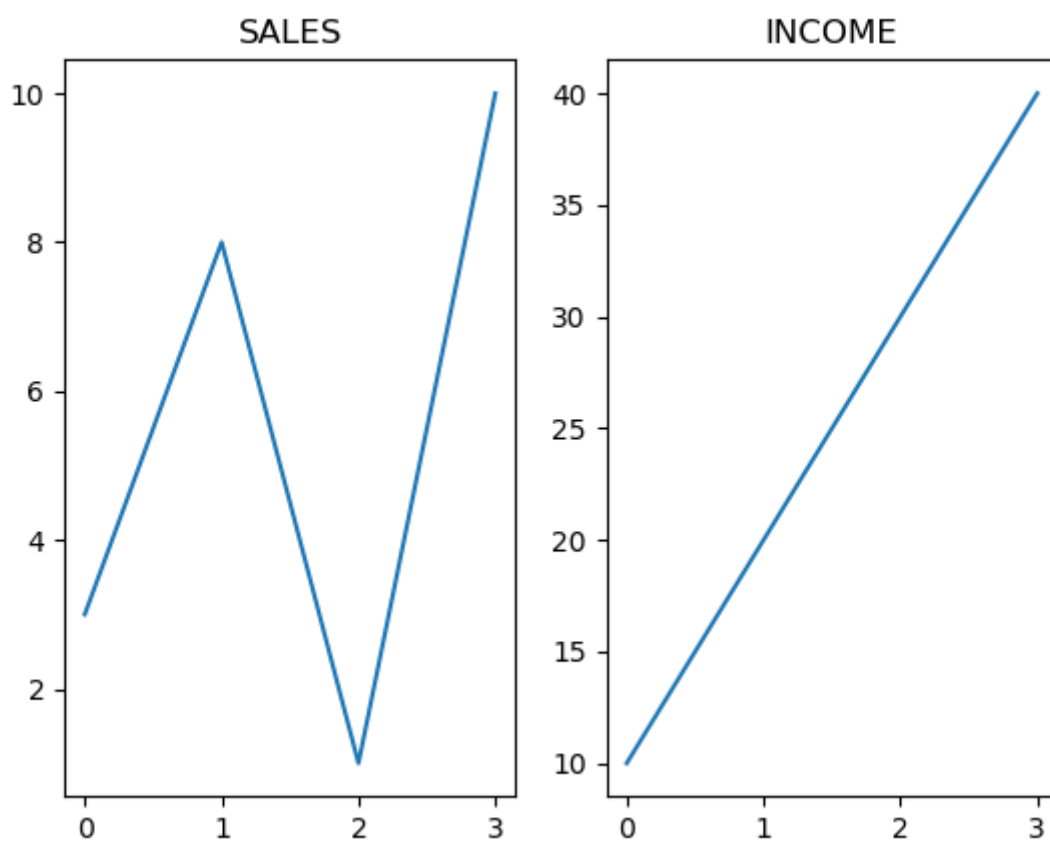
```

```
plt.plot(x,y)
plt.title("SALES")

#plot 2:
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])

plt.subplot(1, 2, 2)
plt.plot(x,y)
plt.title("INCOME")

plt.show()
```



## Super Title

- We can add a title to the entire figure with the `suptitle()` function:

```
import matplotlib.pyplot as plt
import numpy as np

#plot 1:
x = np.array([0, 1, 2, 3])
y = np.array([3, 8, 1, 10])

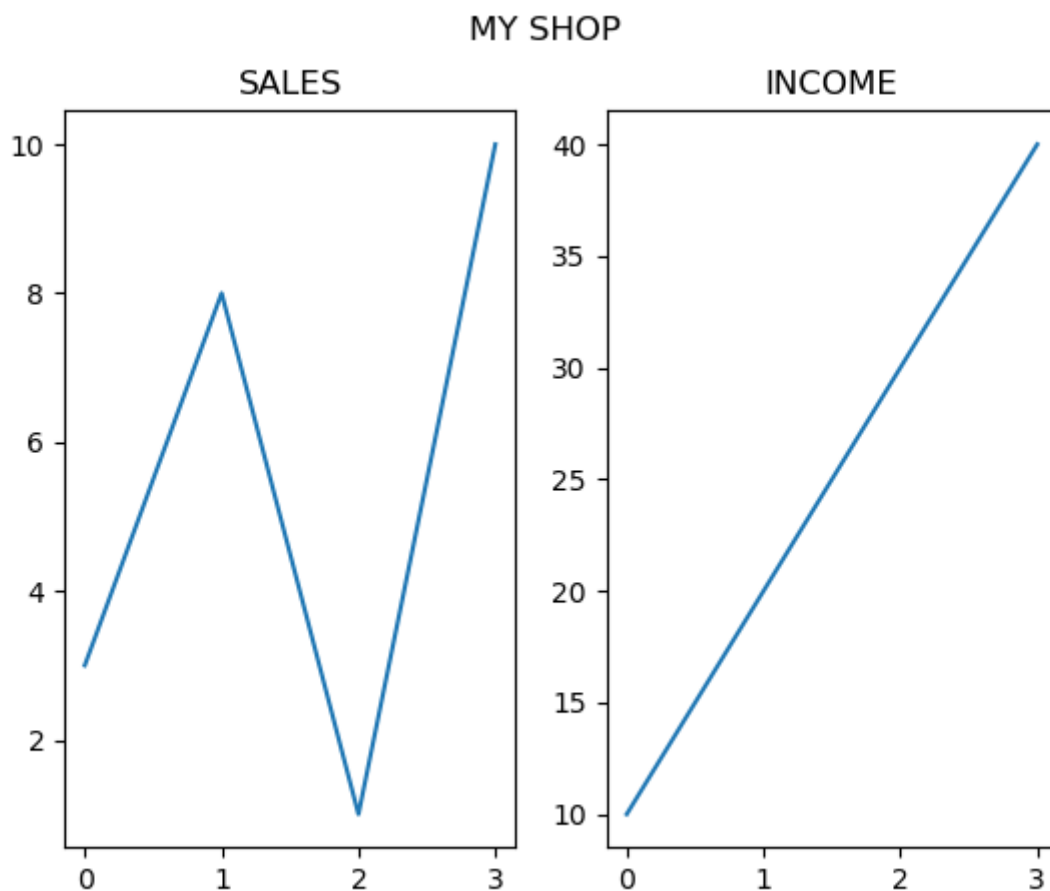
plt.subplot(1, 2, 1)
```

```
plt.plot(x,y)
plt.title("SALES")

#plot 2:
x = np.array([0, 1, 2, 3])
y = np.array([10, 20, 30, 40])

plt.subplot(1, 2, 2)
plt.plot(x,y)
plt.title("INCOME")

plt.suptitle("MY SHOP")
plt.show()
```



## Matplotlib Scatter

- We can use the **scatter()** function to draw a scatter plot.
- The **scatter()** function plots one dot for each observation. It needs two arrays of the same length, one for the values of the x-axis, and one for values on the y-axis:

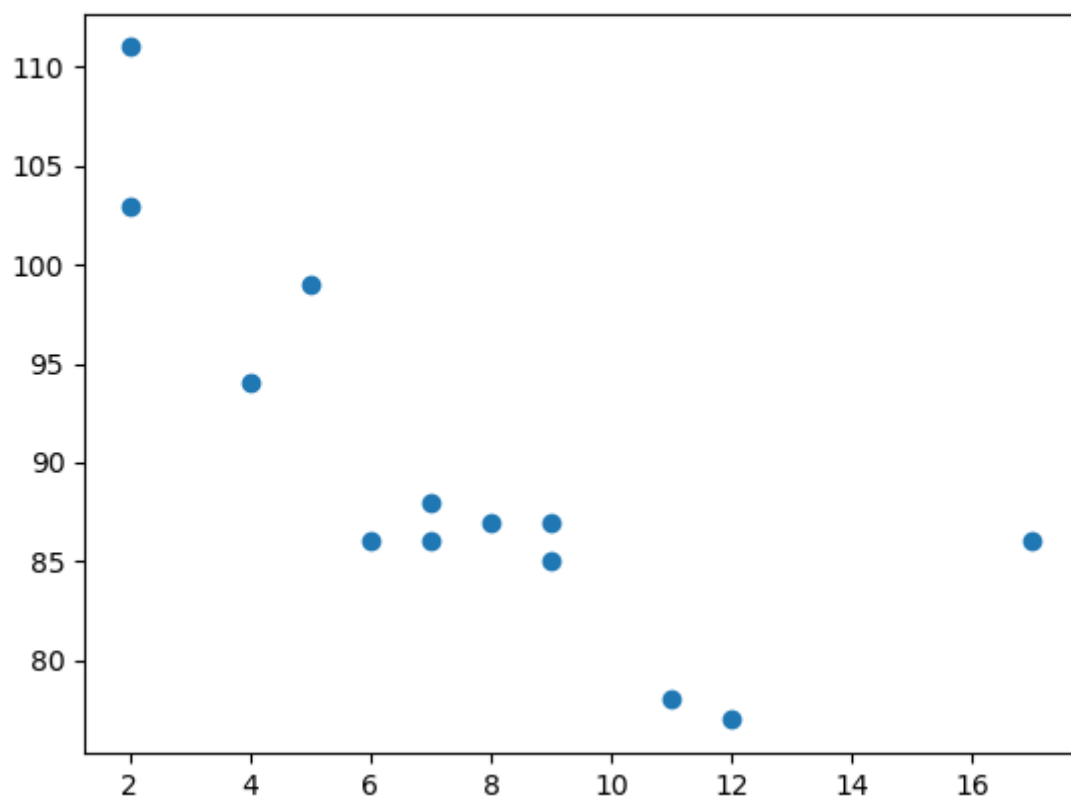
```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5, 7, 8, 7, 2, 17, 2, 9, 4, 11, 12, 9, 6])
```

```
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])

plt.scatter(x, y)
plt.show()
```

- The observation in the example above is the result of 13 cars passing by.
- The **X-axis shows** how **old** the **car** is.
- The **Y-axis shows** the **speed** of the **car** when it passes.
- Are there any relationships between the observations?
- It seems that the newer the car, the faster it drives, but that could be a coincidence, after all we only registered 13 cars.



## Compare Plots

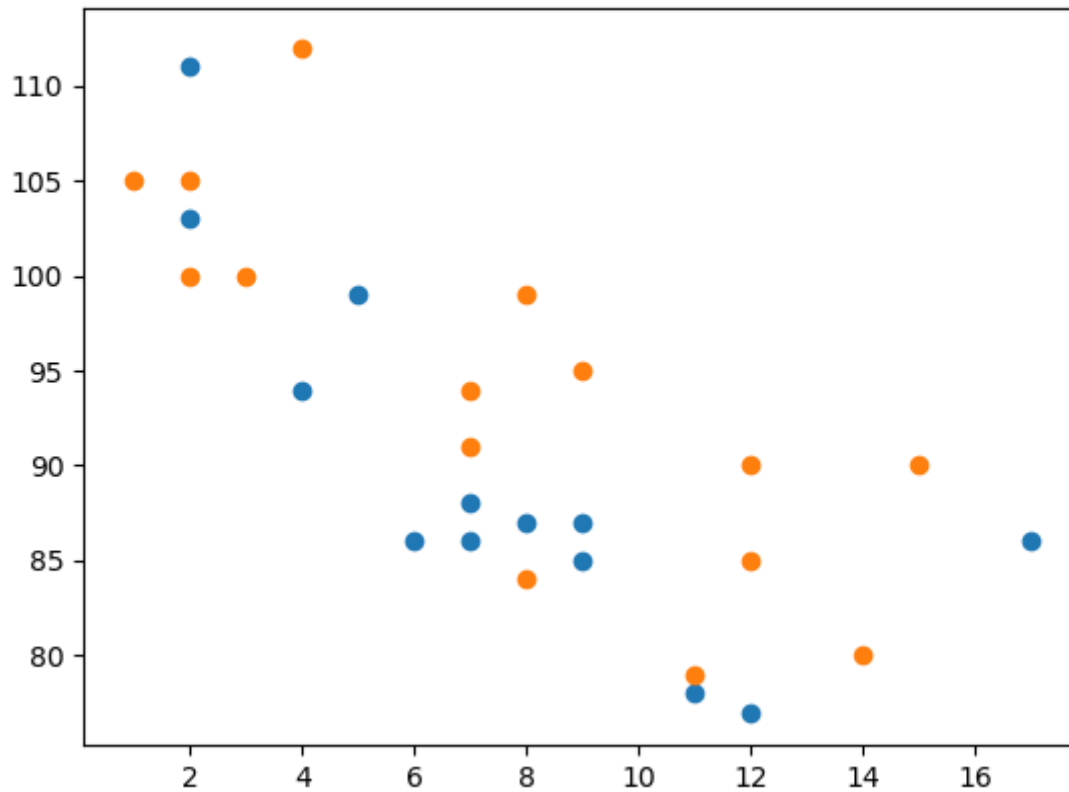
```
import matplotlib.pyplot as plt
import numpy as np

#day one, the age and speed of 13 cars:
x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
plt.scatter(x, y)

#day two, the age and speed of 15 cars:
x = np.array([2,2,8,1,15,8,12,9,7,3,11,4,7,14,12])
y = np.array([100,105,84,105,90,99,90,95,94,100,79,112,91,80,85])
```

```
plt.scatter(x, y)

plt.show()
```



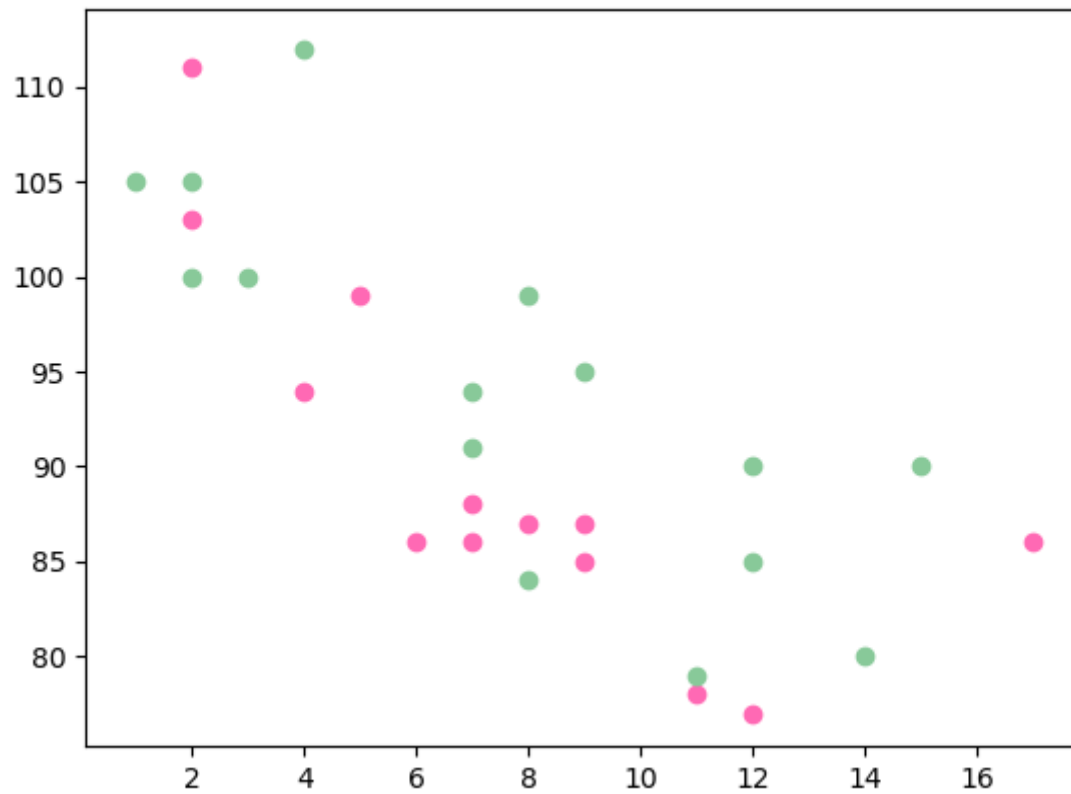
## Change Scatter Colors

```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
plt.scatter(x, y, color = 'hotpink')

x = np.array([2,2,8,1,15,8,12,9,7,3,11,4,7,14,12])
y = np.array([100,105,84,105,90,99,90,95,94,100,79,112,91,80,85])
plt.scatter(x, y, color = '#88c999')

plt.show()
```



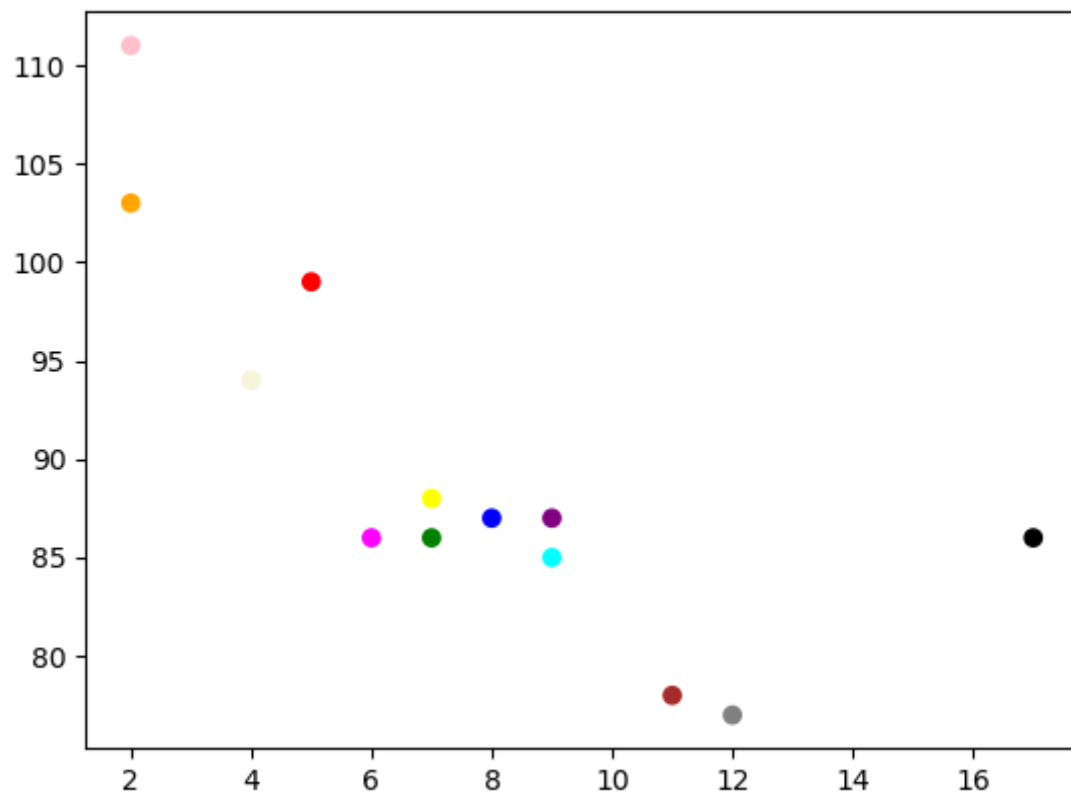
## Color Each Dot

```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
colors =
np.array(["red","green","blue","yellow","pink","black","orange","purple","beige","
brown","gray","cyan","magenta"])

plt.scatter(x, y, c=colors)

plt.show()
```



## How to Use the ColorMap

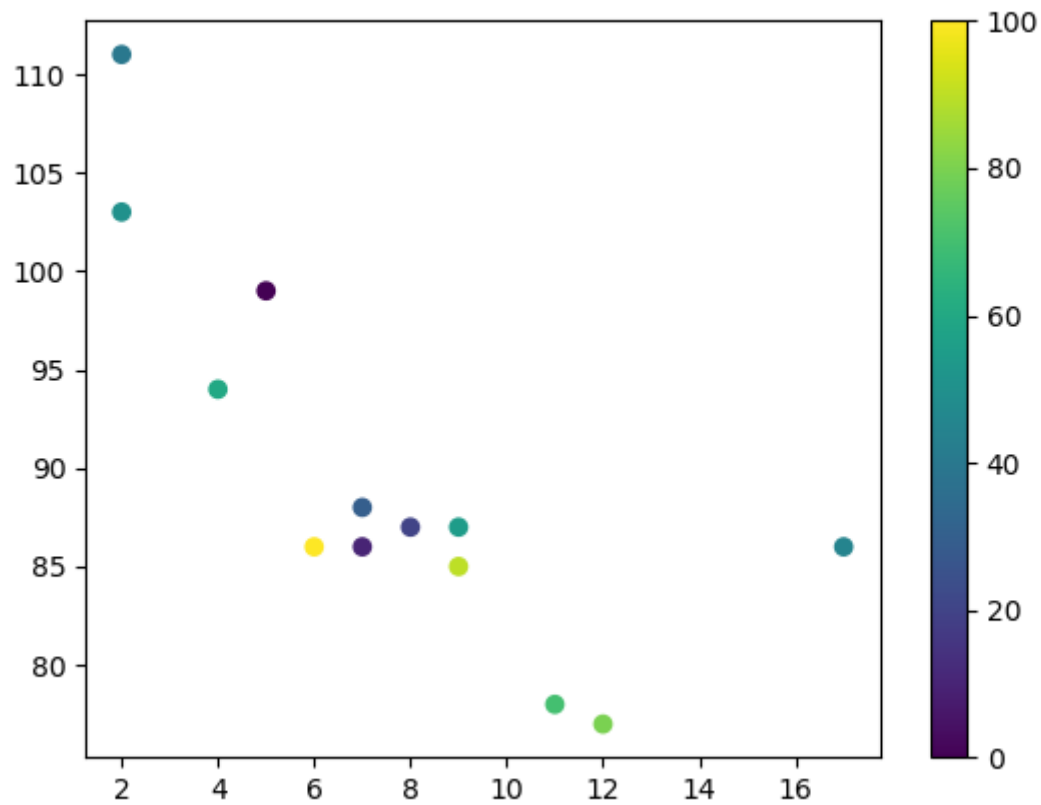
```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
colors = np.array([0, 10, 20, 30, 40, 45, 50, 55, 60, 70, 80, 90, 100])

plt.scatter(x, y, c=colors, cmap='viridis')

plt.colorbar()

plt.show()
```

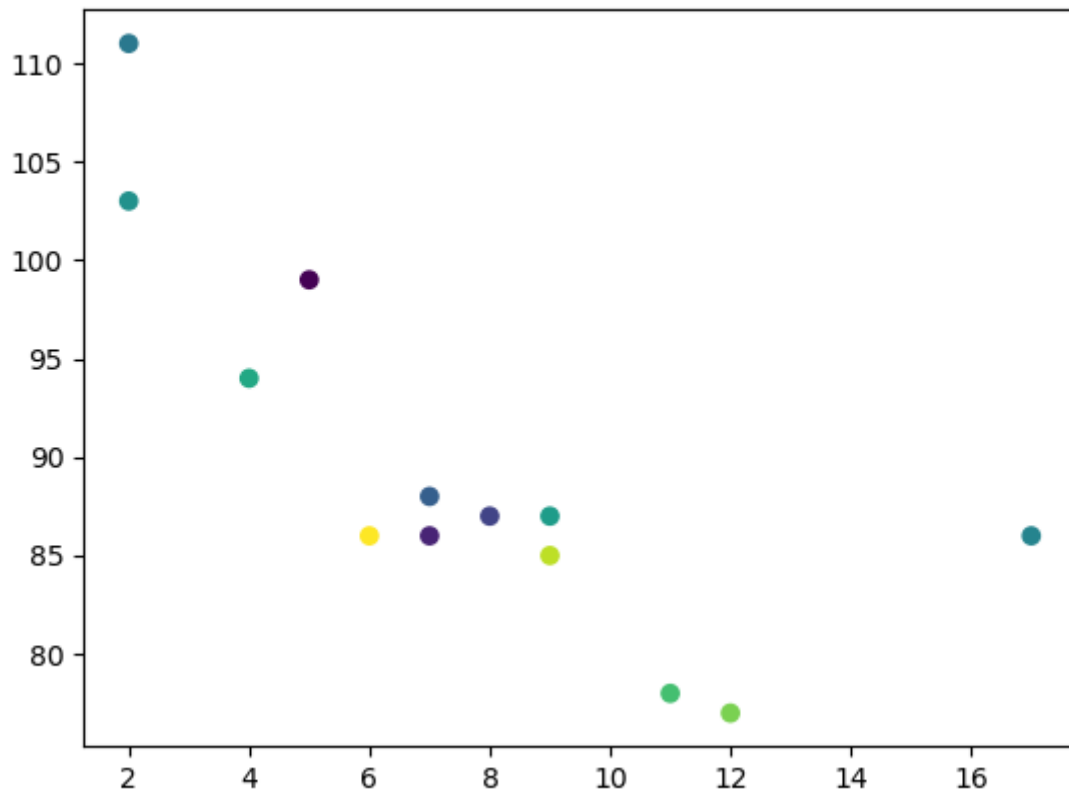


```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
colors = np.array([0, 10, 20, 30, 40, 45, 50, 55, 60, 70, 80, 90, 100])

plt.scatter(x, y, c=colors, cmap='viridis')

plt.show()
```



## Change Scatter Size

- You can change the size of the dots with the **s** argument.

```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
sizes = np.array([20,50,100,200,500,1000,60,90,10,300,600,800,75])

plt.scatter(x, y, s=sizes)

plt.show()
```

## Alpha

- We can adjust the **transparency** of the dots with the alpha argument.

```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
```

```
sizes = np.array([20,50,100,200,500,1000,60,90,10,300,600,800,75])

plt.scatter(x, y, s=sizes, alpha=0.5)

plt.show()
```

## Combine Color Size and Alpha

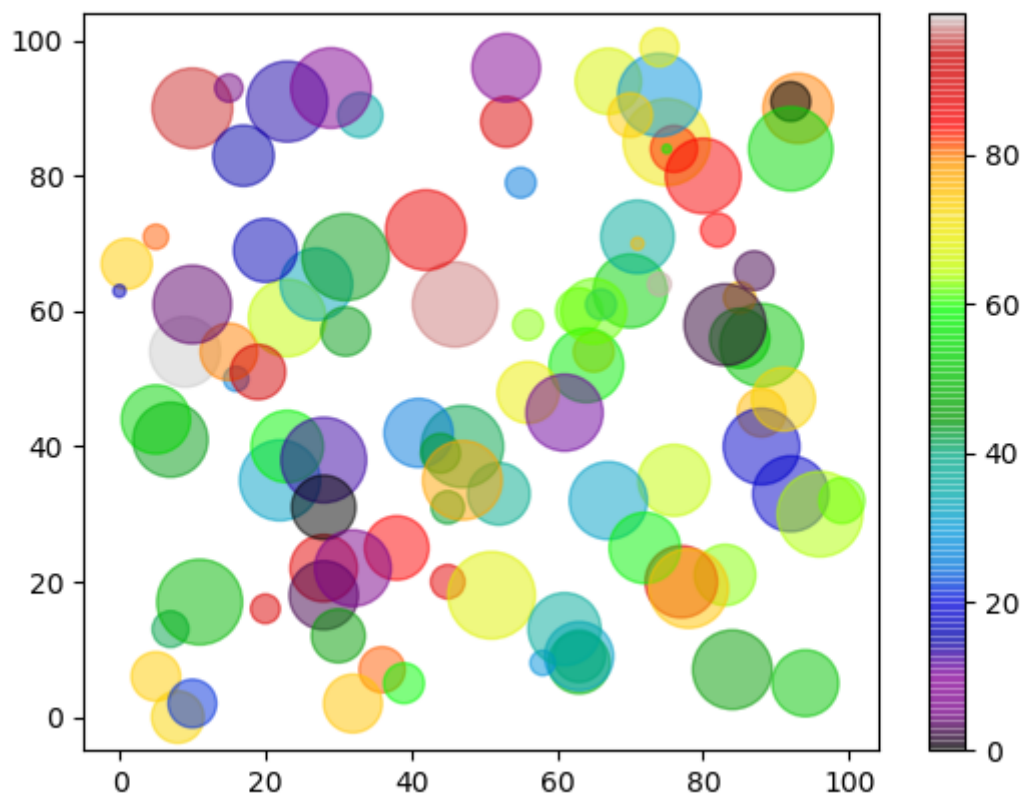
```
import matplotlib.pyplot as plt
import numpy as np

x = np.random.randint(100, size=(100))
y = np.random.randint(100, size=(100))
colors = np.random.randint(100, size=(100))
sizes = 10 * np.random.randint(100, size=(100))

plt.scatter(x, y, c=colors, s=sizes, alpha=0.5, cmap='nipy_spectral')

plt.colorbar()

plt.show()
```



## Matplotlib Bars

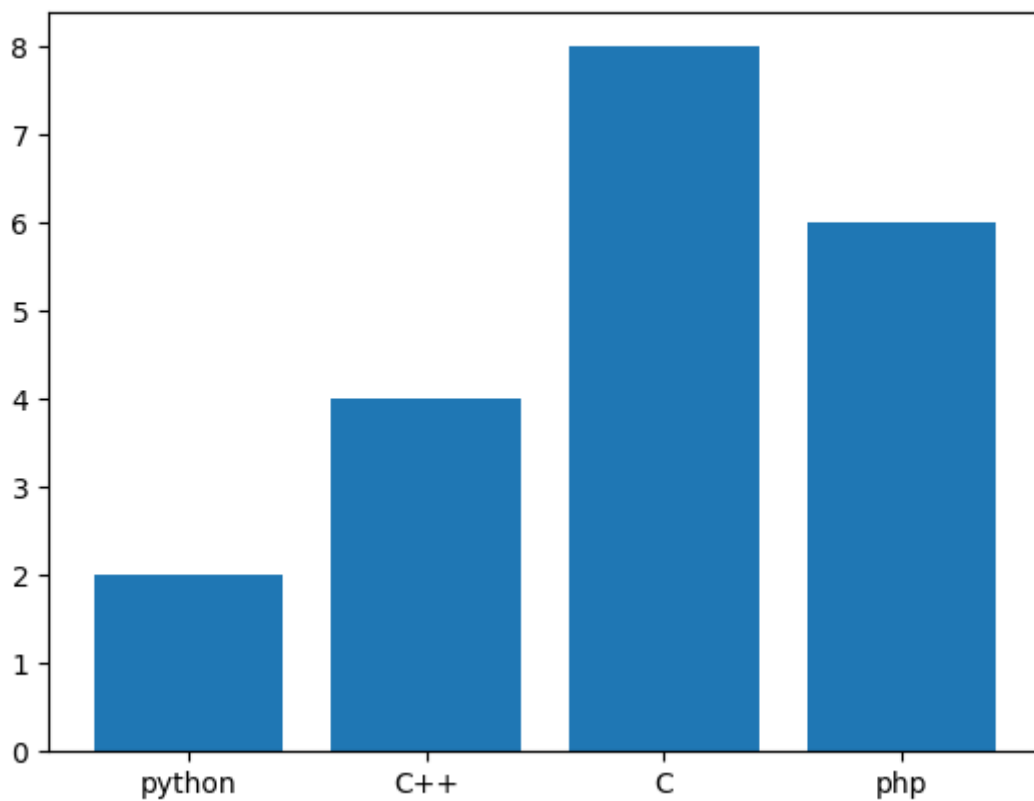
## Creating Bars

- We can use the **bar()** function to draw bar graphs.

```
import matplotlib.pyplot as plt

x=["python","C++","C","php"]
y =[2,4,8,6]

plt.bar(x,y)
plt.show()
```



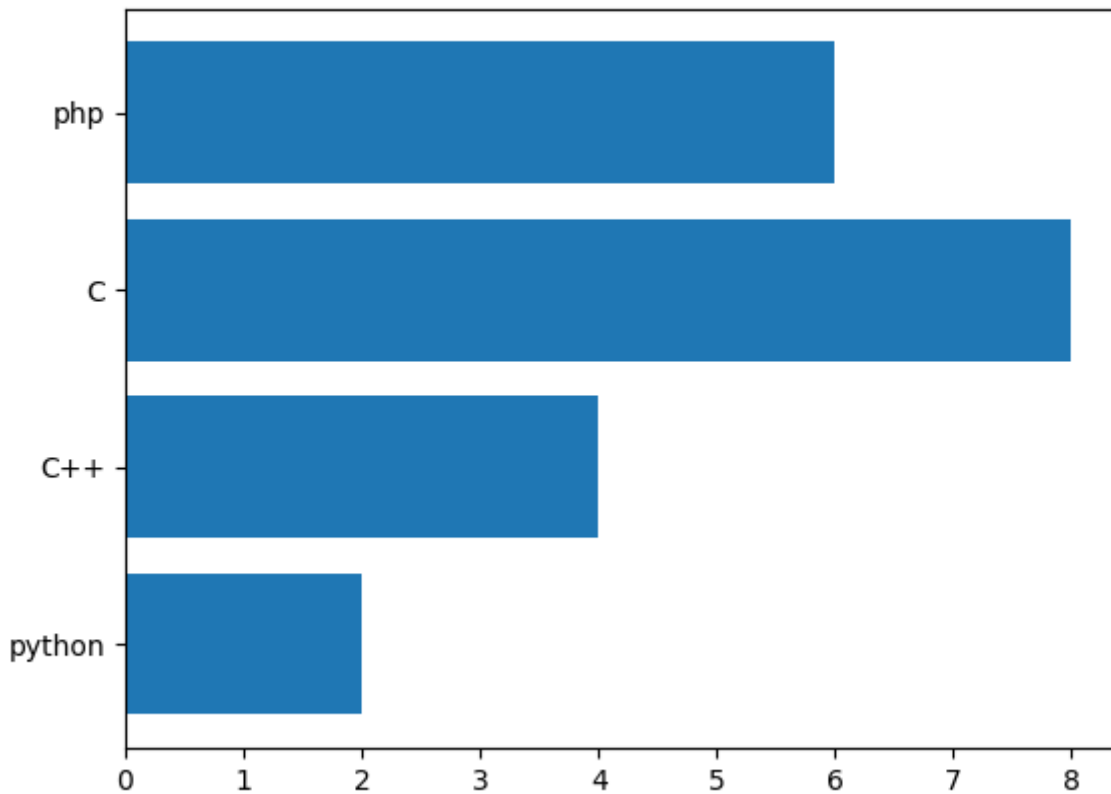
## Horizontal Bars

- If you want the bars to be displayed horizontally instead of vertically, use the **barh()** function:

```
import matplotlib.pyplot as plt

x=["python","C++","C","php"]
y =[2,4,8,6]

plt.barh(x,y)
plt.show()
```



- Bar color for single

```
plt.bar(x,y, color="y")
```

- Multipal color col = ['r','y','g','b'] `plt.bar(x,y,width=0.4, color="col")`
- Bar Width
  - The **bar()** takes the keyword argument **width** to set the width of the bars:

```
plt.bar(x, y, width = 0.1)
```

- The **barh()** takes the keyword argument **height** to set the height of the bars:

```
plt.barh(x, y, height = 0.1)
```

## Bar color

```
# single color  
plt.bar(x,y,width=0.4, color="y")
```

```
# mutipal color
col = ['r','y','g','b']
plt.bar(x,y,width=0.4, color="col")
```

## Align bar chart:- By default center

```
plt.bar(x,y,width=0.4, align="edge")    # for left
plt.bar(x,y,width=0.4, align="center")  # for center
```

## Boundary

```
# Boundary
plt.bar(x,y,width=0.4, edgecolor="r")

# for border width
plt.bar(x,y,width=0.4, edgecolor="r", linewidth="10")

# Boundary line style
plt.bar(x,y,width=0.4, edgecolor="r", linewidth="10", linestyle=":")

# Boundary line style ko halka karna ho to
plt.bar(x,y,width=0.4, edgecolor="r", linewidth="10", linestyle=":", alpha="0.4")
```

```
# bar chart ka label
plt.bar(x,y,width=0.4, edgecolor="r", label="python")
plt.legend()
plt.show()

# two bar graph overlap
import matplotlib.pyplot as plt
import numpy as np

x=["python","C++","C","php"]
y =[8,4,6,7]
z =[4,2,4,3]

c = ['r','y','g','b']

plt.xlabel("language", fontsize="40")
plt.ylabel("no", fontsize="30")
plt.title("mohit", fontsize="20")
#plt.bar(x,y,width=0.4,color="r", label=)
plt.bar(x,y,width=0.4, color="r", label="python1")
plt.bar(x,z,width=0.4, color="y", label="python2")
plt.legend()
```

```
plt.show()

# two bar code without overlap
```

# Matplotlib Pie Charts

---

## Creating Pie Charts

- we can use the **pie()** function to draw pie charts.

```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])

plt.pie(y)
plt.show()
```



## Pie Labels

```
y = np.array([35, 25, 25, 15])  
mylabels = ["Apples", "Bananas", "Cherries", "Dates"]  
  
plt.pie(y, labels = mylabels)  
plt.show()
```

